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REVIEW

Enhanced rock weathering for sustainable carbon capture: advances, challenges, and future prospects in terrestrial ecosystems

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Abstract Enhanced Rock Weathering (ERW) is a carbon dioxide removal (CDR) strategy that accelerates mineral dissolution in terrestrial systems to draw down atmospheric carbon dioxide (CO_2), with the potential for co-benefits across managed and natural ecosystems. This review evaluates ERW's real-world effectiveness through the lens of measurable carbon fluxes and storage permanence. We synthesize evidence across geochemical kinetics, hydrology, mineralogy, particle size distributions, and biological mediation, showing that the wide range of reported field-scale carbon removal rates (approximately 0 to 9 t CO_2 per ha^{-1} yr^{-1}) reflects diverse site-specific environmental conditions and complex geochemical, biological, and hydrological processes that cannot be explained by intrinsic mineral reactivity alone. Widely used proxies, such as total cation dissolutions, soil pH changes, or crop yield gains, are less reliable indicators of field-scale CDR than alkalinity export or cation mass balance. Furthermore, estimates of net CDR should account for both the efficiency of converting weathering products into durable carbonate storage and full lifecycle emissions. Lifecycle assessments consistently identify energy for grinding and logistics as the dominant sources of cost and emissions, implying an optimal particle fineness for

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maximizing net removal. A tiered appraisal of modeling approaches highlights remaining uncertainty across geochemical, ecohydrological, and socioeconomic processes in current models, with plausible 2050 deployment potential estimated at 0.2–4 Gt CO_2 yr^{-1}. We further assess environmental and social risks, including trace-metal accumulation, long-term soil health, and occupational hazards. We conclude by outlining a framework for credible scaling that integrates rigorous monitoring, multi-scale modeling, and precautionary governance. Under these conditions, ERW emerges not as a universal solution, but as a potentially meaningful component of a diversified carbon removal portfolio.

keywords Enhanced rock weathering · Carbon sequestration · Basalt amendment · Monitoring · Reporting · Verification

1 Introduction

The Earth's long-term carbon cycle is closely linked to terrestrial weathering and ocean carbonate mineral formation, processes that act as natural regulators of atmospheric carbon dioxide (CO_2) concentrations (Walker et al. 1981; Berner 1994). Building on these natural regulatory mechanisms, interest has therefore grown in carbon dioxide removal (CDR) approaches, including enhanced rock weathering (ERW), which accelerates naturally occurring mineral weathering to generate alkalinity and support durable atmospheric CO_2 removal (Berner 1992; Ludwig et al. 1999). In this review, we define ERW as a suite of approaches that accelerate carbonation and durable carbon storage in agricultural and natural terrestrial ecosystems by increasing the in situ weathering rate of applied mineral amendments beyond the natural background weathering rate. Within this terrestrial scope, carbonation refers to CO_2-driven mineral dissolution, alkalinity generation, and, where saturation conditions permit, secondary carbonate formation. Crucially, while the fundamental geochemical principles are shared, we distinguish applications in intensively managed croplands from those in less-managed natural or semi-natural ecosystems such as grasslands and forests. Croplands usually permit repeated spreading, shallow incorporation, fertilizer records, yield observations, and control plots, whereas natural ecosystems impose stronger constraints on disturbance, access, baseline attribution, biodiversity safeguards, and hydrologic tracing. We therefore treat these settings as separate interventions requiring tailored measurement, reporting, and verification (MRV) and risk-assessment frameworks rather than as a single deployment category (Beerling et al. 2020; Swoboda et al. 2022; Deng et al. 2023; Levy et al. 2024). Although there is growing interest in carbon storage within synthetic materials such as concrete, these pathways, while valuable, fall outside the terrestrial ERW scope of this review.

With this definition and scope established, early research on ERW began with the recognition that natural rock weathering has substantial potential for large-scale atmospheric CO_2 removal, and that these processes could be intentionally accelerated through engineering (Caldeira 1995; Kojima et al. 1997; Gaillardet et al. 1999). Foundational studies also clarified the geochemical mechanisms that influence rock weathering, establishing a theoretical basis for future research (Matsuoka 1995). Subsequent research on ERW progressed by exploring the potential of different Ca- and Mg-bearing silicate materials, including basaltic, mafic, and ultramafic materials, because their weathering can release alkalinity-generating cations relevant to CO_2 removal. In the early 2000s, studies began to emphasize the significant role of naturally occurring shales and basalts on CO_2 uptake through geochemical weathering, providing a new avenue for understanding the role of different rock types in carbon sequestration (Amiotte Suchet et al. 2003). By 2014, the focus shifted towards optimizing the CO_2 efficiency of enhanced weathering, with researchers highlighting the importance of selecting appropriate source rocks and material comminution techniques (Moosdorf et al. 2014; Liu and Dreybrodt 2015). Towards the end of the last decade, the potential for large-scale CO_2 removal through ERW gained prominence. Notably, Beerling et al. (2018) provided a detailed assessment of the techno-economic potential of implementing ERW on croplands in the UK, including the associated costs and engineering challenges. Beerling et al. (2020) further argued that ERW stands out in the environmental mitigation landscape by presenting a detailed sub-national level analysis of its cost-effectiveness and atmospheric CDR efficacy. Together, these studies helped move ERW from a conceptual weathering-enhancement strategy

toward a deployable CDR approach requiring empirical validation.

Building on this earlier trajectory, research on ERW has evolved markedly, extending to encompass the geochemical efficiency of carbon capture under diverse environmental settings (Deng et al. 2023; Knapp et al. 2023; Ryan et al. 2024; Epihov et al. 2024), broader sustainability considerations (Feng and Hicks 2023; Dupla et al. 2023; Bi et al. 2024), and social equity dimensions (Cox et al. 2022; Sovacool et al. 2023; Low et al. 2024). This broader emphasis positions ERW not only as a geochemical CDR strategy, but also as an ecosystem intervention with potentially significant environmental and societal implications (Taylor et al. 2021). On the environmental side, for example, under appropriate conditions, ERW can improve soil quality (te Pas et al. 2023; Skov et al. 2024), enhance soil fertility (Bi et al. 2024), reduce erosion (Edwards et al. 2017), and promote biodiversity within soil ecosystems (Hicks et al. 2022). On the societal side, studies have highlighted ERW's versatility, showcasing its potential applications in diverse settings from urban farming (Haque et al. 2021) to traditional agricultural landscapes (Swoboda et al. 2022; Buckingham and Henderson 2024). Adjacent research has also explored unconventional ERW pathways integrated into infrastructure (e.g. engineered soils containing silicate minerals) although these applications are treated here primarily as contextual developments rather than core terrestrial ERW pathways (Jorat et al. 2022). Building on these diverse applications, recent scholarship (2024–2025) has increasingly pivoted toward ensuring the holistic viability of terrestrial deployment. The focus has shifted from theoretical potential to rigorous implementation frameworks that connect field quantification, MRV design, lifecycle accounting, and crediting eligibility. This shift is important because carbon-crediting methodologies and registry requirements increasingly determine whether ERW removals can be converted into durable credits, meaning that large-scale deployment cannot rely on agronomic co-benefits alone (Clarkson et al. 2024; Wolf et al. 2024). Parallel efforts have intensified on assessing sustainability performance, supply chain ethics (Oppon et al. 2024a), and long-term environmental safety, such as trace metal accumulation and soil health risks in agricultural systems (Vandeginste et al. 2024; Levy et al. 2024). Recent system-level analyses by Beerling et al. (2025a) and Power et al. (2025) further refine cost estimates and reaction-rate, helping to ground theoretical promises in field and agronomic constraints.

Against this background, while ERW holds promise as a CDR method, the wide range and complexity of factors affecting its long-term effectiveness, scalability, environmental consequences, and economic feasibility introduce substantial uncertainty. This review aims to provide a comprehensive assessment of our current understanding of ERW as a technology, integrating geochemical, ecosystem, and socio-economic perspectives, and highlighting state-of-the-art research directions and key knowledge gaps. We compare theoretical and field-measured CDR and weathering rates, showing order-of-magnitude differences across rock types and deployment contexts, and critically evaluate the measurement approaches (e.g., Brunauer–Emmett–Teller [BET] surface area versus field-effective surface area) and environmental processes (e.g., passivation, leaching rates, and grain-size–dependent dissolution timescales) that may account for these discrepancies. We also assess the integral roles of microbial activity, plant roots, and soil fauna in enhancing rock-powder dissolution, as well as their interactive effects on soil nutrient cycling and terrestrial ecosystem carbon dynamics. In addition, this review provides a comprehensive synthesis of lifecycle, economic, and policy evidence to evaluate the practicality and scalability of ERW. We also compare the strengths and limitations of ERW modeling approaches across spatial scales and lifecycle stages, and further contrast ERW with other CDR strategies to position it within the broader climate-mitigation landscape. Finally, we identify and discuss several critical knowledge gaps. These include the limitations of using state variables, such as soil pH and soil bulk cation concentrations, as direct proxies for carbon removal, highlighting the need for alkalinity- or CO_2 flux-based quantification and for a better understanding of the timescales and uncertainties associated with field-based assessments. We also underscore the importance of linking field-scale CDR effectiveness with Earth system feedbacks and lifecycle optimization; bridging spatially explicit process modeling with integrated assessment frameworks to ensure policy recommendations are based on realistic estimates; and adequately addressing and mitigating environmental risks associated with ERW deployment.

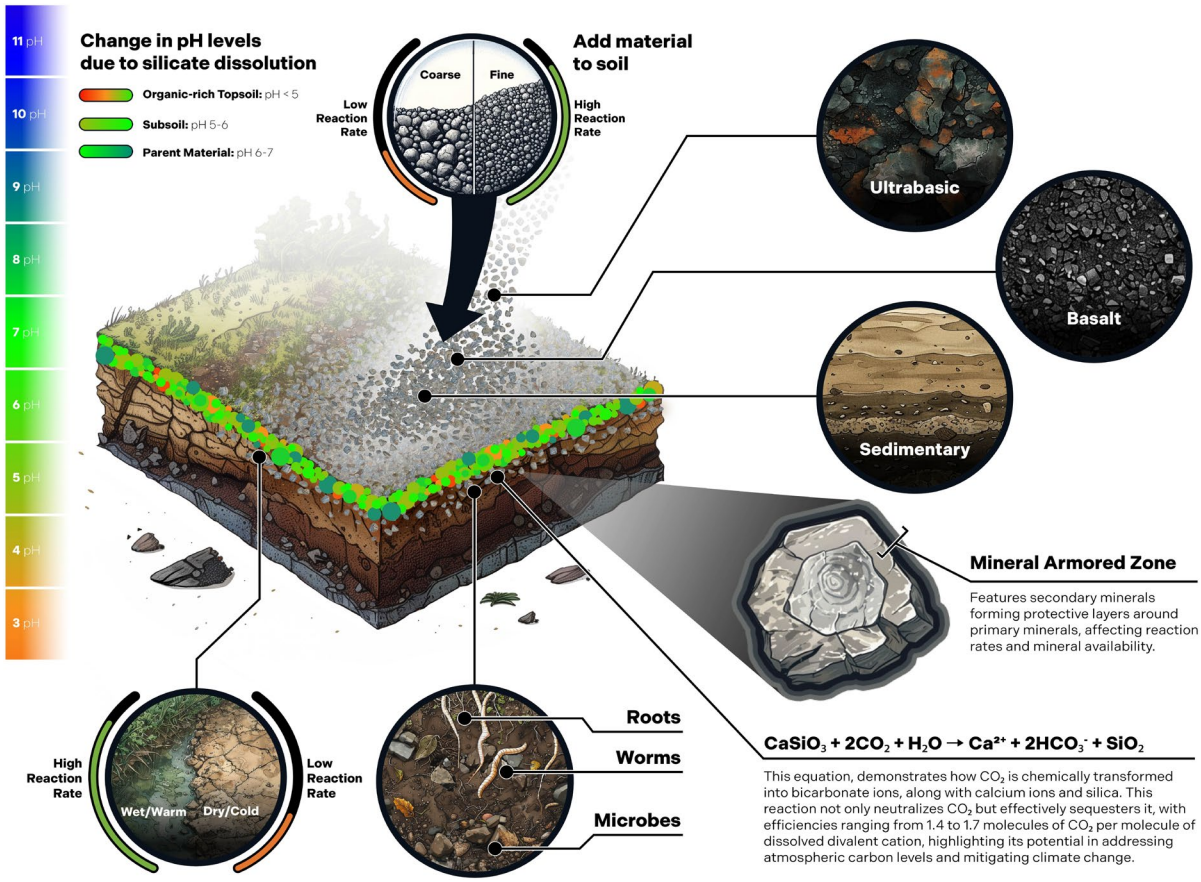


Fig. 1 Geochemical Dynamics and Biophysical Interactions in Enhanced Rock Weathering. This infographic depicts how different rock types (e.g., ultrabasic, basaltic, and sedimentary materials) and particle sizes are incorporated into soils to accelerate mineral weathering. These reactions release alkalinity-generating cations and convert CO_2 into dissolved bicarbonate and carbonate species that may be transported to durable sinks or stored as secondary carbonates. The figure also illustrates pH gradients across soil layers, the influence of microbes, plant roots, and soil fauna on mineral breakdown, and the role of particle size in controlling reactive surface area. Secondary mineral formation and surface armoring are shown as mechanisms that can reduce mineral availability and lower field-effective reaction rates over time

2 Foundational geochemical dynamics of ERW

Figure 1 summarizes the basic reaction pathways of ERW, alongside biophysical interactions and grain size effects. Silicate dissolution rates are affected by the surface area of the rock powder, soil pH, hydrological, and biological conditions (Sect. 2.1). These factors co-evolve with the dissolution process over time (Sect. 2.2–2.5, Sect. 3.1–3.3), resulting in no single "rate constant". Additionally, the alkalinity produced by silicate dissolution must either be transported away from the point of origin into durable ocean sinks, or be retained in pedogenic sinks, to result in durable CDR (Vakilifard et al. 2021; Zhang et al. 2025). Therefore, the CDR potential of ERW cannot be simply inferred from the stoichiometry of the source rock's mineralogy alone.

2.1 Reaction mechanisms

In the context of ERW for CDR, the primary mechanism considered is the weathering of silicate minerals (Hartmann et al. 2013). This process generates

alkalinity that is either transported from soil through rivers to the ocean, where it is ultimately stored over geologic timescales, or retained in soils as solid carbonate minerals where saturation conditions allow. Although carbonate weathering can also consume CO_2 over short timescales (e.g., $<10^5$ years; (Zhang et al. 2022)), its long-term net effect on atmospheric CO_2 is approximately neutral because carbonate precipitation or ocean carbonate compensation ultimately re-releases an equivalent amount of CO_2. Therefore, in this review, we focus on silicate-derived alkalinity as the primary pathway for durable ERW CDR.

Silicate minerals are naturally occurring compounds composed of silicon and oxygen, typically combined with metal cations. Their dissolution is the fundamental geochemical process underlying ERW. This reaction involves silicate minerals, CO_2, and water, resulting in the release of alkali and alkaline earth cations, primarily Ca^{2+}, Mg^{2+}, K^+, and Na^+:

$$\text{Silicates} + CO_2 + H_2O \rightarrow \text{Cations} + HCO_3^- + SiO_2 \quad (1)$$

Over decadal timescales, this reaction increases the alkalinity of soil and aquatic environments through the formation of bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}) ions (Hartmann et al. 2013). These dissolved inorganic carbon species may be transported through drainage and river networks to the ocean, where they can be stored over geologic timescales. The released Ca^{2+} and Mg^{2+} can further accelerate the formation of pedogenic carbonate minerals (e.g. calcite), constituting a second CDR pathway. This results in the net reaction:

$$\text{Silicates} + CO_2 + H_2O \rightarrow \text{Carbonates} + SiO_2 \quad (2)$$

For both pathways, removal is only durable when dissolved or solid products are retained or transported to sinks without being offset by CO_2 degassing, carbonate compensation, or lifecycle emissions (Hartmann et al. 2013).

Since the silicate dissolution reaction is the starting point of ERW as well as the natural weathering cycle, the rate of silicate dissolution (r, mol s^{-1}) has been extensively studied (see Brantley 2008 for a review). The rate law is typically formulated as a function of chemical affinity (the distance from equilibrium), specific surface area, solution pH, temperature (T), and mineral type (Palandri and Kharaka 2004):

$$r = -m \cdot SSA \cdot \sum_I \left\{ k_i \exp\left[-\frac{E_i}{R}\left(\frac{1}{T} - \frac{1}{298.15K}\right)\right] a_i^{n_i}(1-\Omega^{p_i})^{q_i} \right\}$$

$$\Omega = \frac{Q}{k_{eq}}$$

$$Q = \frac{\Pi_m [A_m]^{c_m}}{\Pi_n [A_n]^{c_n}}$$

where the definitions are:

$i \in \{H^+, H_2O, OH^-\}$—different weathering agents under acid, neutral, and basic conditions. m—the mass of the mineral, g. SSA—specific surface area, $m^2\ g^{-1}$. k_i—dissolution rate constants at 298.15 K, mol $m^{-2}\ s^{-1}$. E_i—Arrhenius activation energy of the reaction, J mol^{-1}. T—temperature, K. R—natural gas constant, 8.31 J K^{-1} mol^{-1}. a_i—activity of the aqueous species i, with $a_{H_2O} = 1$ and $a_{H^+}a_{OH^-} = 10^{-14}$. n_i—reaction order with respect to species i, $n_{H_2O} = 1$. Ω—the mineral saturation index. p_i, q_i—empirical dimensionless constants, generally set to 1 due to lack of information. Q—activity product. k_{eq}—equilibrium constant for the dissolution reaction. $[A_m]$—concentration or activity of the aqueous species m on the right-hand side of the dissolution reaction equation. $[A_n]$—concentration or activity of the aqueous species n on the left-hand side of the dissolution reaction equation. c_m, c_n—stoichiometric coefficients of the dissolution reaction equation.

This widely used dissolution rate law captures dependence on chemical affinity, pH, temperature, and surface area (Brantley 2008), but it is important to recognize that this rate law is an abstraction of the actual dissolution process. In reality, defect-controlled reaction hotspots, incongruent dissolution and the formation of amorphous silica that promotes surface passivation, as well as temperature effects that can increase intrinsic rates while simultaneously driving solutions closer to equilibrium, can all cause observed dissolution rates to diverge from the theoretical constants presented in Table 1. Additionally, the values reported in Table 1 arise from different reactors, background electrolytes, distances from equilibrium, and area metrics, which complicates direct comparison. As a result, cross- study comparisons require explicit normalization for pH, saturation state, temperature, grain size, and the surface-area metric used (Brantley 2008). Furthermore,

Table 1 Dissolution rate constants (Log k, mol m^{-2} s^{-1}) and Arrhenius activation energies (E, kJ mol^{-1}) of selected land ERW minerals under acidic and neutral mechanisms

Classification of silicate minerals		Formula	Acid mechanism				Neutral mechanism			Study[a]
			log k	E	n	pH	log k	E	pH	
Orthosilicates	Fayalite (Olivine)	Fe_2SiO_4	−4.80	94.4	1.000	–	–	94.4	–	(Renforth et al. 2015)
	Forsterite (Olivine)	Mg_2SiO_4	−6.85	67.2	0.470	1.03–8.71	−10.64	79.0	9.30–12.06	
Phyllosilicates	Lizardite	$Mg_3Si_2O_5(OH)_4$	−5.70	75.5	0.800	–	−12.40	56.6	–	(Taylor et al. 2016)
Inosilicates	Wollastonite	$CaSiO_3$	−5.37	54.7	0.400	–	−8.88	54.7	–	(Jariwala et al. 2022)
	Tremolite	$Ca_2Mg_5Si_8O_{22}(OH)_2$	−8.40	18.9	0.700	–	−10.60	94.4	–	(Paulo et al. 2021)
	Diopside	$CaMgSi_2O_6$	−6.36	96.1	0.710	–	−11.11	40.6	–	(Lewis et al. 2021)
Tectosilicates	Anorthite	$CaAl_2Si_2O_8$	−3.50	16.6	1.411	–	−9.12	17.8	–	(Baek et al. 2023)
	Albite	$NaAlSi_3O_8$	−10.2	65.0	0.457	1.3–4.0	−12.6	69.8	5.6–8.2	(te Pas et al. 2023)
	K feldspar	$KAlSi_3O_8$	−10.1	51.7	0.500	4–5.91	−12.4	38.0	6–7	(Xiao et al. 2016)

a: The references listed in the "Study" column indicate studies in which the corresponding minerals were applied or investigated in ERW-related or dissolution studies. The kinetic parameters (log k, E, n) and the acid/neutral reaction mechanisms are derived from the internally consistent mineral dissolution rate compilations of Palandri and Kharaka (2004) and the updated database of Oelkers and Addassi (2025)

laboratory-derived weathering rates should be treated as condition-specific constraints rather than directly transferable field-scale constants because they do not fully capture the complexity of geochemical, hydrological, and biological processes under natural field conditions. Nevertheless, the qualitative trend that less polymerized silicate frameworks dissolve more rapidly than tectosilicates remains robust.

To address the need for long-term predictability, recent analyses contrast the decadal-scale dissolution trajectories of basalt and olivine. Under representative soil conditions (pH 5–6), olivine (forsterite) typically exhibits a 'fast-start' trajectory, yielding rapid initial alkalinity fluxes that may decline sharply within 3–5 years due to surface passivation (formation of Si-rich layers) or depletion of the fine fraction (Power et al. 2025). In contrast, basalt, being a multi-mineral assemblage, shows a more complex, staggered release profile. While its bulk dissolution is slower, the heterogeneity of its mineral phases (e.g., plagioclase vs. pyroxene) creates a more sustained, albeit lower-magnitude, alkalinity flux over a 10-year horizon, provided that secondary clay formation (e.g., smectites) does not occlude reactive surfaces (Calabrese et al. 2022; Beerling et al. 2025a). Thus, while olivine provides higher theoretical carbon-removal potential per unit mass, basalt may offer greater kinetic stability over decadal deployment cycles. A practical implication is that field verification should ensure that short-term cation releases in soil solution are consistent with the stoichiometry of the target dissolving mineral, and that long-term monitoring may be required to verify whether early alkalinity fluxes persist. Under these conditions, Table 1 is best used not as a catalogue of fixed numerical rates, but as a prior for constraining field-effective dissolution rates once hydrologic contact conditions and silica-driven feedbacks have been characterized.

2.2 Alkalinity modulation and pH dynamics

Because silicate dissolution alters both alkalinity and acidity, pH is one of the most observed but also most easily misinterpreted ERW indicators. The pH responses in Table 2 span from approximately 3.6–8.5 across weathering conditions, lithologies, particle sizes, application rates, and exposure durations. This range confirms that ERW can rapidly shift soil acidity, particularly under fine particles, larger inputs, and

Table 2 Soil pH responses to ERW across rock type, location, particle size, application rate, and exposure duration

Rock types	Location	Soil type	Temp	Particle Size (μm)	Application rate (g/m^2)	Duration (month)	pH change	Study
Basalt	Iceland	Andosol	7 °C	10–62	500–800	7	4.4–6	(Linke et al. 2024)
Basalt	Australia	Oxisols	Room temp	< 150	2500	3	4.75–5	(Gillman et al. 2002)
Basalt	Malaysia	Oxisols	26–28°C	< 250	500–8000	3–15	3.9–5.1	(Anda et al. 2009)
Basalt	Malaysia	Oxisols	26–28°C	< 250	500–2000	4–24	4.0–5.3	(Anda et al. 2013)
Basalt	UK	clay-loam agricultural soil	N/A	50–150	10,000	2	6.6–6.7	(Kelland et al. 2020)
Basalt	UK	Sandy clay-loam	N/A	< 4000	1886	8	6.47–6.76	(Skov et al. 2024)
Dunite	Belgium	N/A	N/A	< 43.5	22,000	12	6.6–8.5	(Amann et al. 2020)
Dunite	Belgium	N/A	N/A	< 1020	22,000	12	6.6–8.0	
Olivine	Netherlands	Podzols	15−20 °C	< 200	163	8	4.89–4.99	(ten Berge et al. 2012a, b)
Olivine	Netherlands	Podzols	15−20 °C	< 200	815	8	4.89–5.10	
Olivine	Netherlands	Podzols	15−20 °C	< 200	4,070	8	4.89–5.34	
Olivine	Netherlands	Podzols	15−20 °C	< 200	20,400	8	4.89–5.96	
Olivine	Denmark	Podzols	20−25 °C	20	5000	3	3.56–4.69	(Dietzen et al. 2018)
Olivine	Denmark	Podzols	20−25 °C	20	1000	3	3.56–5.18	
Wollastonite	USA	Podzols	6 °C	16	344	12	3.88–4.39	(Johnson et al. 2014)
Steel slag	China	Ultisol	25 °C	< 1000	5–26	9	5.82–6.68	(Ning et al. 2016)

longer treatments (Gillman et al. 2002; Anda et al. 2009, 2013; ten Berge et al. 2012a, b; Johnson et al. 2014; Ning et al. 2016; Dietzen et al. 2018; Amann et al. 2020; Kelland et al. 2020; Linke et al. 2024). Nevertheless, these pH responses should be interpreted as evidence of acid neutralization and reaction progress rather than as standalone measures of CDR. A short-term pH rise can reflect exchange-site occupation, proton consumption, or increasing saturation state without demonstrating exported alkalinity or durable carbonate storage. This distinction is important because pH elevation can also increase the saturation state (Ω) of soil solution and dampen further proton-promoted dissolution. Accordingly, pH trajectories are most informative when interpreted together with bicarbonate concentrations, Ca^{2+} and Mg^{2+} balances, major-ion chemistry, discharge or residence time, and baseline controls (Almaraz et al. 2022; Clarkson et al. 2024). Without that closure, pH shifts represent localized reaction potential rather than realized sequestration.

The agronomic narrative is equally contingent. Enhanced pH can improve nutrient availability and microbial activity and can raise yields in some contexts, but the magnitude and persistence of these benefits depend on soil buffering capacity, background acid loads, and management practices (Liu et al. 2021; Swoboda et al. 2022; Guo et al. 2023; Wood et al. 2023). Consequently, studies that invoke co-benefits as part of the value proposition should co-report time-resolved pH, alkalinity export, and nutrient budgets to distinguish durable removal from transient soil conditioning.

Two interpretive priorities emerge for Fig. 1 and Table 2. First, distinguish clearly between pH adjustment and alkalinity generation, and attach comparable hydrologic and ionic data to each pH observation. Second, emphasize the feedback whereby pH elevation can slow proton-driven dissolution, since this sets a natural trajectory of early rapid response followed by kinetic dampening.

2.3 Particle size and surface area

Because accessible surface area mediates these pH and alkalinity responses, finer particles increase

geometric surface area per unit mass, and mesocosm evidence and global modeling support higher near-term alkalinity generation under finer distributions and larger pH shifts for the same rock input (Taylor et al. 2016; Strefler et al. 2018; Amann et al. 2020; Beerling et al. 2020; Baek et al. 2023) Even so, three systematic deviations from the idealized picture must be accounted for. First, surface area does not scale simply with sieve size when micro-porosity, defects, and aggregation dominate; BET surface area, measured by gas adsorption, and geometric estimates can diverge. Power et al. (2025) recently highlighted a critical systemic error in modeling: using BET (gas adsorption) surface area often overestimates field reaction rates because internal micropores accessible to gas are often inaccessible to soil water or are quickly clogged by secondary precipitates. However, geometric surface area is also not a direct measure of field-reactive area; it is typically estimated from particle-size distributions, particle density, and simplified grain-shape assumptions, and can underestimate rough, fractured, porous, or aggregated grains. Thus, BET, geometric, and field-effective surface areas should be reported as distinct quantities and, where possible, calibrated against measured cation release and alkalinity fluxes. Second, the field-available surface area decays over time because selective erosion removes fines, while passivation and pit coalescence reduce active site density (Pokrovsky and Schott 2000; Tuğrul 2004; Brantley 2008; Nguyen et al. 2016; White et al. 2017; Swoboda et al. 2022; Calabrese et al. 2022). Third, energy requirements for comminution grow rapidly as target size decreases, and lifecycle emissions can overwhelm the alkalinity gains if grinding is pushed too far (Rinder and von Hagke 2021) (Sect. 4.2). The operational implication is that net carbon removal per unit energy has an interior optimum rather than increasing monotonically with finer particle size. Identifying this optimum requires paired experiments that report both alkalinity fluxes and energy use, along with measurements of time-varying particle size distributions in situ.

These particle-size effects also propagate into model uncertainty. Model comparisons reveal that assumptions about grain size distributions, surface-area normalization, shrinking-core behavior, and mineral specificity drive large differences in predicted removal at the same application rate (Taylor et al. 2016; Strefler et al. 2018; Beerling et al. 2020; Baek et al. 2023). These assumptions are only weakly constrained by field data and constitute a significant source of model structural uncertainty, especially when BET-derived, geometric, and field-effective surface areas are compared without a shared normalization framework.

2.4 Mineral armoring and coupled reactions

As reactive surfaces evolve during weathering, dissolution may be further inhibited by mineral armoring resulting from incongruous dissolution or secondary mineral precipitation. Passivation layers formed through the preferential depletion of cations near the mineral surface or through coupled dissolution–precipitation reactions further reduce the access of reactive fluids to fresh mineral surfaces. While models have traditionally focused on the kinetic suppression caused by amorphous Si-rich layers (Daval et al. 2011; Amann et al. 2020), recent evidence highlights that secondary carbonate precipitation itself can drive surface armoring. Notably, Weber et al. (2023) observed that Mg-carbonate precipitation can act as a primary passivating agent on Mg-rich reactive surfaces, effectively inhibiting further reaction. This behavior contrasts with calcium-rich phases (e.g., wollastonite and other Ca-rich materials), where secondary carbonate precipitates often fail to form coherent blocking layers, allowing dissolution to persist.

Reaction–diffusion treatments can replicate these diverse passivation behaviors, yet parameterization remains challenging in heterogeneous soils and is seldom included in large-scale ERW assessments. Omitting these mechanisms—whether driven by silica or secondary carbonates—likely biases removal upward when saturation states increase over time (Emmanuel 2022; Eufrasio et al. 2022; Baek et al. 2023; Weber et al. 2023). Passivation does not necessarily preclude meaningful ERW, but it means that high weathering extents such as $> 20\%$, $> 50\%$, or near-complete dissolution should not be assumed from initial rates alone. They require sustained hydrologic flushing, appropriate particle-size distributions, reactive feedstock selection, and long-term verification of cation release and alkalinity export.

Hydrologic conditions further modulate passivation. Under high rainfall, pore waters dilute, supersaturation declines, and passivation can weaken, which

favors dissolution. Under drying, evaporation and limited flushing raise saturation ratios and encourage precipitation, which enhances armoring. Because reaction direction depends on saturation state, the balance between dissolution and precipitation varies with the residence time distribution and with transport pathways rather than being a fixed material constant (Brantley 2008; Roelandt et al. 2010). For this reason, long-term field studies should co-monitor dissolved Si, cation release stoichiometry, and the morphology of secondary precipitates to relate silica exposure histories to effective rate decay and to constrain model representations of passivation.

2.5 Environmental conditions

Because ERW occurs in heterogeneous soils rather than well-mixed reactors, and because durable carbon removal depends on coupled terrestrial and aquatic processes, either of which may become the rate-limiting step, weathering conditions alone are meaningful but insufficient predictors of carbon removal. Natural weathering accelerates in warm, wet environments, and modeling studies often reproduce higher ERW potentials in the tropics and subtropics (Brady and Carroll 1994; Levenson et al. 2017; Goll et al. 2021; Baek et al. 2023). Yet high rainfall can shorten fluid residence times and truncate contact between pore waters and reactive surfaces, while riverine processing can partially negate upland alkalinity gains before they reach marine sinks (Klemme et al. 2022; Zhang et al. 2022). Therefore, environment-driven "fast kinetics" does not automatically translate into "more removal." The controlling variables are the joint distribution of residence times and saturation states along the flow path.

Soil properties and land management exert first-order influence on this joint distribution. Structure, connectivity, and cation exchange capacity determine infiltration, water holding, soil CO_2 profiles, and pore water chemistry, all of which reshape field-effective rates and the allocation of products among exchange sites, dissolved loads, and secondary phases (Maher and Chamberlain 2014; Johnson et al. 2014; Deng et al. 2023). Tillage, cover cropping, and harvest remove or redistribute cations and alter mixing depths, which in turn modify alkalinity budgets and transport (Beerling et al. 2020; Deng et al. 2023). Consequently, realistic assessments should be built around mass balance and alkalinity flux rather than background descriptions of setting. A reasonable working expectation is that, at comparable weather conditions, soil structural metrics that control residence time will explain more variance in net removal than climate variables alone. This expectation is testable with paired, cross-soil trials that combine tracer-informed hydrology with alkalinity export measurements.

3 Ecosystem synergies of ERW

Biota modulate ERW not as a uniform accelerator but as a set of interacting controls on reaction thermodynamics, transport, and element budgets. Figure 1 places microbes, roots, and fauna alongside mineral surfaces and hydrologic pathways. Synergistic benefits only occur when biological processes increase verifiable alkalinity export, maintain acceptable risk profiles, and avoid countervailing fluxes such as respiration-driven CO_2 release or nutrient losses. Accordingly, the sections below evaluate where evidence supports durable gains, where mechanistic claims rest on laboratory surrogates, and where field-scale flux closure remains unproven.

3.1 Soil microbial processes

Microbial metabolism can increase mineral dissolution through acid production, biofilm-mediated attachment, and ligand-promoted complexation, and several studies document catalytic enhancement of rock weathering under controlled conditions (Blackmore et al. 2018; Pastore et al. 2022). However, much of the most detailed microbiological evidence comes from carbonate or mixed-lithology contexts, which limits direct transfer to silicate-dominated ERW unless alkalinity production and export are shown. Moreover, community composition and function vary with moisture, temperature, and N to P ratios, so climatic gradients confound simple extrapolation from single-site experiments (Xu et al. 2018). Consequently, claims that microbial processes "accelerate ERW" require co-measurement of porewater alkalinity, cation stoichiometry, and residence times, together with attribution of CO_2 sources, since microbial respiration can increase pore-space pCO_2 while not necessarily contributing to net atmospheric

drawdown. In practical terms, biofilm formation and organic acid secretion may increase near-surface dissolution yet also raise dissolved silica, which can promote surface passivation and lower effective rates over time; studies that report only early-time kinetics risk overstating sustained removal. Recent work by Epihov et al. (2024) demonstrates that microbial siderophores can significantly accelerate silicate dissolution through iron chelation. However, this biological enhancement operates within strict geochemical limits. As noted by Levy et al. (2024), microbial enhancement of weathering must be evaluated alongside trace-metal release; olivine-rich ultramafic feedstocks are a classic case in which Ni and Cr mobilization can affect soil microbial communities and risk assessment. Therefore, studies that report only early-time kinetics risk overstating sustained removal if they ignore these biotic, geochemical, and toxicity-related feedback loops.

3.2 Plant roots

Plant roots and mycorrhizal fungi provide a second biological pathway through which ERW can be modified in the field. Plant roots and mycorrhizal fungi amplify weathering by elevating pore-space CO_2, secreting organic acids, and intensifying exchange between solids and solutions, and these interactions are central to realistic ERW scenarios in managed and unmanaged systems (Dontsova et al. 2020; Swoboda et al. 2022; Deng et al. 2023). Organic ligands chelate metal centers and mobilize cations, thereby advancing dissolution beyond what proton attack achieves alone (Jones et al. 2009; Quirk et al. 2012). However, these biological synergies are expressed differently across land-use settings. In croplands, annual rooting patterns, fertilizer additions, tillage or incorporation depth, and harvest exports directly shape cation budgets and alkalinity export. In forests and grasslands, deeper perennial root systems, lower levels of disturbance, the absence of fertilizer input and biomass harvesting, and greater biodiversity make ERW deployment and monitoring fundamentally different from field applications in agricultural systems. Thus, managed and natural systems should be assessed as separate interventions: cropland studies can pair agronomic records with aqueous fluxes, whereas natural-system pilots require stronger hydrologic baselines, tracer support, and safeguards for soil biota and habitat disturbance (Swoboda et al. 2022; Deng et al. 2023; Bi et al. 2024; Levy et al. 2024; Beerling et al. 2025a). In practical terms, croplands prioritize agronomic nutrient cycling, yield, and harvest exports, whereas natural ecosystems prioritize deep-profile inorganic carbon storage, hydrologic stability, and biodiversity protection. The same proton-coupled transport process that mobilizes cations also uptakes them into biomass; if Ca and Mg accumulate in plant tissues and are removed at harvest, the exportable alkalinity in soil might be curtailed, which affects net removal claims in croplands. On the other hand, roots modify redox microenvironments by transporting oxygen and shaping electron acceptor distributions. The resulting cycling of Fe between reduced and oxidized states may promote the reprecipitation of Si-rich or Fe–Si secondary phases on dissolving grain surfaces, potentially reducing the reactive surface area (Quirk et al. 2012).Therefore, studies that report biotic acceleration of ERW should also report matched cation budgets, biomass export, and evidence for downstream alkalinity transport. Enhanced dissolution at the root-mineral interface or increased bulk cation concentration in the soil does not guarantee field-scale removal without mass-balance closure.

3.3 Soil fauna

Beyond microbial and root-mediated effects, soil fauna influence ERW primarily through physical restructuring of the medium rather than through direct catalysis. Burrowing, casting, and ingestion mix rock powders into reactive zones, fragment particles, and alter moisture and aeration, which can increase contact between porewaters and mineral surfaces (Osler and Sommerkorn 2007; Xu Guo-Liang and Kuster 2012; Morford et al. 2016; Pinto-Figueroa et al. 2019; Tan et al. 2021). However, these same processes can also transport fines out of the targeted horizons and increase hydrologic connectivity, shortening residence times and potentially reducing the fraction of alkalinity that is retained long enough to be exported in solution. Evidence for fauna-driven acceleration is strongest at microcosm and plot scales and over short durations; few studies couple bioturbation metrics with time-resolved alkalinity fluxes and particle size distributions in situ. Until such data are available, the most defensible representation of fauna in ERW assessments is as modifiers of mixing depth,

hydraulic residence time distributions, and reactive surface exposure, rather than as independent "boosters" of intrinsic dissolution kinetics. Crucially, however, this interaction is bidirectional. While fauna facilitate weathering through bioturbation, they are also the first receptors of potential toxicity. Reviewers have rightly pointed out the risk of antagonistic impacts: Levy et al. (2024) and Dupla et al. (2024) warn that the accumulation of trace metals (e.g., Ni, Cr) from ultramafic rocks can reach thresholds that inhibit earthworm activity or reduce population density. Consequently, monitoring soil fauna health is not merely a metric of biodiversity, but a necessary safeguard to verify that the physical-acceleration function of ERW remains functional over long-term deployment.

3.4 Soil nutrient cycles and terrestrial ecosystem carbon cycles

The biological controls discussed above interact with soil nutrient cycling and terrestrial carbon storage. Nutrient feedback offers a plausible path to durable ERW gains, but the direction and magnitude of the effects depend on soil chemistry and management. Increased pH from silicate dissolution can alter nitrification and denitrification pathways and has been proposed to lower N_2O and NO emissions in some settings, yet the sign of the response depends on moisture regimes, substrate availability, and microsite dynamics, so global-scale reductions should be treated as contingent rather than universal (Val Martin et al. 2023). Phosphorus dynamics provide a second axis of interaction. ERW can raise P availability through direct apatite dissolution when this mineral is present in the basalt feedstock, and through enhanced mobilization of organic P mediated by microbial and root processes, which can shift plant uptake strategies toward greater mycorrhizal reliance and improve nutrient-use efficiency in P-limited systems (Goll et al. 2021; Bi et al. 2024). At the same time, not all silicate rocks provide substantial P, and pH shifts can precipitate P into less available forms in some soils, so lithology and background chemistry determine whether ERW delivers an agronomic dividend or a neutral outcome (Manning 2018; Swoboda et al. 2022; te Pas et al. 2023).

Carbon cycle responses extend beyond dissolved inorganic carbon. Release of Ca, Mg, and Si can stabilize soil organic carbon through mineral–organic associations and aggregate formation, potentially increasing the fraction of soil organic carbon (SOC) in more persistent pools, although documentable gains depend on the balance between enhanced inputs and any priming of decomposition following pH changes (Hartmann et al. 2013; Beerling et al. 2020; Xu et al. 2024). Recent validation studies challenge the assumption of uniform SOC benefits. Wolf et al. (2024) emphasize that changes in soil pH and microbial activity can trigger the priming effect, potentially accelerating the decomposition of existing SOC pools. Therefore, robust claims about ecosystem synergies must explicitly deduct these potential hidden carbon costs. As highlighted by Beerling et al. (2025a), a verifiable carbon removal credit requires a comprehensive net ecosystem carbon budget that accounts for both inorganic sequestration and any antagonistic losses in organic carbon stocks. Because inorganic carbon formation and organic carbon stabilization have different permanence, monitoring frameworks should separately account for soil inorganic carbon and SOC, report mineral-associated organic matter fractions, and avoid conflating transient pH effects with long-term storage (Manning et al. 2024). Productivity responses add another layer. Field work indicates that ERW can raise net primary production in some systems, including bioenergy crops such as miscanthus, yet higher biomass does not automatically translate into greater net removal if harvest exports base cations or if hydrologic losses counterbalance added inputs (Kantola et al. 2023). Finally, model-based upper-bound estimates of carbon drawdown in P-limited regions, including values approaching 2 kg CO_2 m^{-2} yr^{-1} at application rates of 5 kg m^{-2}, are informative for potential but hinge on assumptions that remain to be verified across soils and weather conditions (Goll et al. 2021). For these reasons, robust claims about ecosystem synergies should be conditioned on co-reported alkalinity export, nutrient budgets, trace gas fluxes, and SOC fractionation, rather than on single proxies such as pH or short-term growth responses.

4 Evaluating ERW: data, lifecycle, economics, and policy considerations

4.1 Data, observations and experiments

After establishing the mechanistic controls on ERW, the next question is how reliably these processes can be measured in experiments and field deployments. Laboratory kinetics for silicate dissolution are well established, yet pilot and field evidence for ERW spans more than four orders of magnitude, which directly propagates into uncertainty in carbon removal estimates (White and Brantley 1995; White 2008). Table 3 makes clear that this spread is not random noise. The compilation spans croplands, grasslands, and forested catchments, and studies quantify different phases, including exchange sites, porewaters, plant uptake, secondary minerals, and adsorbed pools, under distinct hydrologic regimes, mineralogy, and soils. As a result, cross-study comparisons are only meaningful when measurements are anchored in mass balance with clear boundaries. A critical limitation in some observational data is not pH monitoring itself, but the use of isolated state variables or concentration changes without cation balance closure. pH and porewater alkalinity are valuable diagnostic measurements, but CDR quantification requires linking them to cation budgets, discharge or residence time, baseline alkalinity level, and field condition uncertainty (Holzer et al. 2023). While Fig. 2 illustrates a dependency of CDR on pH, recent rigorous reviews by Clarkson et al. (2024) argue that pH shifts are insufficient proxies for carbon removal. They emphasize that without quantifying the 'alkalinity leak'—the actual bicarbonate flux leaving the root zone—observed changes may simply reflect transient cation exchange rather than permanent sequestration. Furthermore, the disparity between lab and field-measured dissolution rates is being re-evaluated. Power et al. (2025) suggest that widely used surface area metrics (e.g., BET) systematically overestimate the reactive surface available in field conditions. Consequently, the weathering rate gaps shown in Fig. 2 are not merely evidence of kinetic suppression but likely also reflect artifacts of initial parameterization. Future experiments must therefore shift from reporting soil solid-phase changes (which are noisy and prone to heterogeneity) to high-frequency aqueous monitoring that can close the mass balance (Wolf et al. 2024). At the same time, flux-based MRV must be designed with sampling intensity and laboratory cost in mind: alkalinity titration, porewater chemistry, isotope analyses, and repeated solid-phase measurements cannot all be applied at the same frequency across large deployments. A practical strategy is therefore tiered, using lower-cost measurements such as pH, electrical conductivity, soil moisture, and discharge for screening and temporal interpolation, while using alkalinity flux and selected tracers to calibrate, verify, and audit the removal estimate (Stubbs et al. 2022; Calabrese et al. 2022; Wolf et al. 2024).

Even with these caveats, two patterns are consistent with first principles and with Table 1 across both agricultural fields and natural landscapes. First, mineral identity matters. Wollastonite applications repeatedly report higher removal and faster apparent weathering, which is congruent with its rapid dissolution under soil pH conditions. Second, pH exerts a strong control. Figure 2 shows that field-normalized rates for olivine vary by more than four orders of magnitude between pH about 3.5 and pH near 8, a spread that exceeds laboratory projections when fitted to standard rate laws. This divergence is expected if selective leaching and secondary precipitation increase with rising dissolved silica and if hydrologic contact times are insufficient to offset passivation. In that case, the relevant comparison is not between laboratory lines and field points but between field points with similar saturation states and residence times.

Site heterogeneity further complicates inference in both managed and unmanaged systems. The Malaysian oil palm study illustrates how background practices and legacy amendments can mask a silicate signal. In that setting, widespread carbonate fertilization raised baseline weathering and alkalinity export, leaving little room for a detectable increment from silicate additions even under warm and wet conditions that should otherwise favor reaction (Larkin et al. 2022). Because such interactions are common in working landscapes, studies that lack controls on background alkalinity inputs risk conflating treatment effects with management history. The same problem can arise from the feedstock itself: carbonate impurities in basaltic materials, sedimentary rocks, skarns, or alkaline residues can dissolve rapidly and produce early alkalinity or cation pulses that are not equivalent to silicate-derived durable CDR. As deployment scales up, these confounders will likely widen uncertainty

Table 3 Pilot- and field-scale observations of ERW

Site/experiment description	Duration	Soil characterization	Feed stock characterization	Experiment condition	Measured phases	Total CO_2 removal (t CO_2/ha over duration)	Weathering extent[a]	Study
Planted mesocosm (PVC pots of 10 L, with top diameter of 25 cm)	32 wk (0.62 yr)	pH: 4.79 CEC[b] (cmol/kg): 19 Texture: sandy Source: Wageningen, Netherlands	Olivine SSA (N_2, BET)[c]: 4.8 m^2/g Particle size: P_{73} = 0.05 mm Application rate: 0.163 kg/m^2 (low); 20.4 kg/m^2 (high)	Olivine powder was fully mixed with soil; Natural rainfall; 400 ml water was supplied on top every 14 d; Duration: 32 weeks	Plant; Exchangeable	0.28 (low); 2.58 (high)	0.148 (low); 1.1×10^{-2} (high)	(ten Berge et al. 2012a, b)
Flow-through reactor on soil cores (0.1×1 m)	5 mo (0.42 yr)	Calcareous soil Source: North Oxfordshire, United Kingdom	Olivine SSA (BET): 3.04 m^2/g Particle size: P_{65} = 0.212 mm Application rate: 12.7 kg/m^2	Olivine powder was applied to top 20 cm layer; Constant flow at 16.8 m/yr Temperature: 19 °C Duration: 5 months	Leachate; Exchangeable; Secondary carbonate	0.54	6.6×10^{-3}	(Renforth et al. 2015)
Incubation experiment (110 ml bottles)	3 mo (0.25 yr)	pH: 3.55 Texture: sandy Source: Nørholm Hede, Denmark	Olivine (Forsterite) Particle size: P_{50} = 0.02 mm Application rate: 1 kg/m^2 (low), 5 kg/m^2 (high)	Olivine powder was fully mixed with soil; De-ionized water was added to maintain soil moisture; Temperature: 20–25 °C Duration: 3 months	Exchangeable	4.15 (high); 3.13 (low)	7.1×10^{-2} (high); 0.267 (low)	(Dietzen et al. 2018)
Planted mesocosm (rooftop pots of 9 L, with top diameter of 24 cm)	8 wk (0.15 yr)	pH: 4.94 Source: Ontario, Canada	Wollastonite SSA (N_2, BET): 20.28 m^2/g Particle size: P_{90} = 0.026 mm Application rate: 22 kg/m^2	Wollastonite powder was fully mixed with soil; Natural rainfall Duration: 8 weeks	Secondary carbonate	39.3	0.74	(Haque et al. 2019a)

Table 3 (continued)

Site/experiment description	Duration	Soil characterization	Feed stock characterization	Experiment condition	Measured phases	Total CO_2 removal (t CO_2/ha over duration)	Weathering extent[a]	Study
Planted mesocosm (length: 26 cm, diameter: 46 cm)	730 d (1 yr reported)	pH (KCl): 6.6, 7.8 CEC (cmol/kg): 8.6 Texture: loamsand Source: Belgium	Olivine (dunite) from Almklovdalen peridotite complex SSA (N_2, BET): 9.5 m^2/g (fine), 1.1 m^2/g (coarse) Particle size: P_{80} = 0.04 mm (fine), 1.02 mm (coarse) Application rate: 22 kg/m^2	Olivine powder was applied to top 11 cm layer; Irrigated daily and weekly; Mean infiltration rate: 800 mm/yr	Leachate	0.049 (fine); 0.023 (coarse)	9.9×10^{-5} (fine); 4.6×10^{-5} (coarse)	(Amann et al. 2020)
Planted mesocosm (length: 50 cm, diameter: 16 cm)	120 d (0.33 yr)	pH (H_2O): 6.6 CEC (cmol/kg): 25.4 Texture: loamsand Source: Leicestershire, United Kingdom	Columbia River Basalt SSA (N_2, BET): 7.35 m^2/g Particle size: P_{80} = 1.25 mm Application rate: 10 kg/m^2	Basalt powder was applied to top 25 cm layer; Irrigated every 5 d; Mean infiltration rate: 767 mm/yr Temperature: ~25 °C Duration: 120 d	Leachate; Plant; Exchangeable; Secondary carbonate	0.78	1.6×10^{-2}	(Kelland et al. 2020)
Soil core (0.1×1 m)	5 wk (0.10 yr)	Calcareous, loamy soil Source: North Oxfordshire, United Kingdom	Columbia River Basalt SSA (N_2, BET): 16.3 m^2/g Particle size: P_{80} = 1.25 mm Application rate: 10 kg/m^2	Basalt powder was applied to top 13 cm layer; Natural rainfall (infiltration rate: 57 mm/yr) Duration: 5 weeks	Leachate (the cores were flushed by 330 ml of rainwater before sampling)	0.001	6×10^{-5}	(Buckingham et al. 2022)
Planted mesocosm (length: 100 cm, diameter: 50 cm)	99 d (0.27 yr)	pH: 7.72 Texture: loamy Source: Belgium	DURUBAS basalt SSA (BET): 9.2 m^2/g Particle size: P_{75} = 0.25 mm Application rate: 5 kg/m^2	Basalt powder was applied to top 20 cm layer; watered regularly with tap water (infiltration rate: 143 mm/yr)	Leachate; Plant; Exchangeable; Secondary carbonate	The detected difference is smaller than soil heterogeneity		(Vienne et al. 2022)

Table 3 (continued)

Site/experiment description	Duration	Soil characterization	Feed stock characterization	Experiment condition	Measured phases	Total CO_2 removal (t CO_2/ha over duration)	Weathering extent[a]	Study
Hubbard Brook Experimental Forest (11.8 ha)	1–11 yr	pH: 3.89 Texture: sandy loam Location: New Hampshire, USA	Wollastonite SSA (BET): 1.6 m^2/g Particle size: P_{50} = 0.016 mm Application rate: 0.345 kg/m^2	Wollastonite powder (pelletized to 1 to 3 mm) was applied by helicopter; MAP [d]: 140 cm Annual discharge: 87 cm MAT [e]: 6 °C Duration: 11 y	Leachate; Plant; Exchangeable	>1.24 (1 yr, river bank); > 1.76 t (11 yr, whole watershed)	> 0.635 (1 yr, river bank); > 0.924 (11 yr, whole watershed)	(Peters et al. 2004) (Green et al. 2013) (Johnson et al. 2014)
Farm (18 ha potato field)	5 mo (0.42 yr)	pH: 5.92 Texture: Sandy loam Location: Alliston, Canada	Wollastonite SSA (N_2, BET): 2.07 m^2/g Particle size: P_{50} = 0.014 mm Application rate: 0.124 kg/m^2	Applied using a lime spreader Irrigated Duration: 5 months	Secondary carbonate	0.32	0.68	(Haque et al. 2020a)
Farm (2 ha soybean field)	20 wk (0.38 yr)	pH: 6.59 Texture: Sandy loam Location: Woodstock, Canada	Wollastonite SSA (N_2, BET): 3.48 m^2/g Particle size: P_{50} = 0.008 mm Application rate: 0.5 kg/m^2	Applied using a lime spreader Not irrigated Duration: 20 weeks	Secondary carbonate	0.40	0.21	
Oil palm plantation (0.6–1.8 ha)	3 yr	pH: 5–5.8 Location: Sabah, Malaysia	Pleistocene magmatic-arc volcanic deposits SSA (N_2, BET): 2.73 m^2/g Particle size: P_{50} = 1.77 mm Application rate: 5 kg/m^2	MAP: ~2000 mm MAT: 28 °C	Leachate; Exchangeable; Secondary carbonate	No significant difference was detected		(Larkin et al. 2022)

Table 3 (continued)

Site/experiment description	Duration	Soil characterization	Feed stock characterization	Experiment condition	Measured phases	Total CO_2 removal (t CO_2/ha over duration)	Weathering extent[a]	Study
Farm (3 × 20 m plots)	3 yr	pH (H_2O): 5.8; pH ($CaCl_2$): 5 CEC (cmol/kg): 7.16 Texture: clay-loam Source: Vojens, Denmark	Ilulialik rock flour SSA (N_2, BET): 19.6 m^2/g Particle size: P_{50} = 0.0026 mm Application rate: 5 kg/m^2	Ilulialik rock flour was applied to top 8 cm layer by harrowing; MAP: 897 mm MAT: 9.4 °C Duration: 3 y	Exchangeable	0.72	7.9×10^{-2}	(Dietzen and Rosing 2023)
Miscanthus plots (0.7–3.8 ha)	5 yr	Texture: silt loam Location: Illinois, USA	Blue Ridge basalt SSA (N_2, BET): 1.02 m^2/g Particle size: P_{80} = 0.276 mm Application rate: 5 kg/m^2	Applied using a lime spreader MAP: 1051 mm MAT: 11 °C Duration: 5 y Yearly application for 3 (miscanthus) and 4 years (maize/soybean)	The difference in total elemental composition in solid phases Plant	43.0	0.29	(Kantola et al. 2023)
Maize/soybean plots (0.7–3.8 ha)	5 yr					18.5	0.09	
Corn/Soybean Farm (Midwest US)	3 yr	pH: Initially acidic Location: Illinois/Midwest, USA	Steel Slag vs. Basalt Rate: 100 t/ha	Conventional agriculture; Comparison of two feedstocks	Porewater pH;Ca-saturation	Steel Slag: Clear CDR evidence (alkalinity ↑) Basalt: No significant change	Steel slag weathered significantly; Basalt did not	(Maxbauer et al. 2025))
Vineyard (Temperate field)	3 yr (1000 days)	pH: Neutral to slightly acidic Location: Switzerland	Basalt Application rate: 20 t/ha (2 kg/m^2)	Applied to surface;Natural rainfall	Porewater;Solid carbonates	~0.3 (0.1 t/ha/yr) (10–30 × lower than models)	Low (Na release was 3 × higher, but C capture limited)	(Dupla et al. 2025)
Sugarcane Farm (Tropical acidic soil)	4 yr	pH: 5.3 (Acidic) Location: Queensland, Australia	BasaltApplication rate: 50 t/ha/yr (High rate)	High rainfall (Tropical);Acidic soil driven by nitric/organic acids	Leachate (HCO_3^- flux)	0.10 (0.026 t/ha/yr) (Negligible net CDR)	Significant weathering occurred, but acidity source was not H_2CO_3	(Holden et al. 2024)

[a] Weathering extent is quantified as the ratio of the mass of minerals that have undergone weathering to the total initial mass of minerals present, over the course of the experiment. [b] CEC: cation exchange capacity. [c] specific surface area (SSA) measured via N_2 adsorption and calculated using the Brunauer–Emmett–Teller (BET) method. [d] MAP: mean annual precipitation. [e] MAT: mean annual temperature

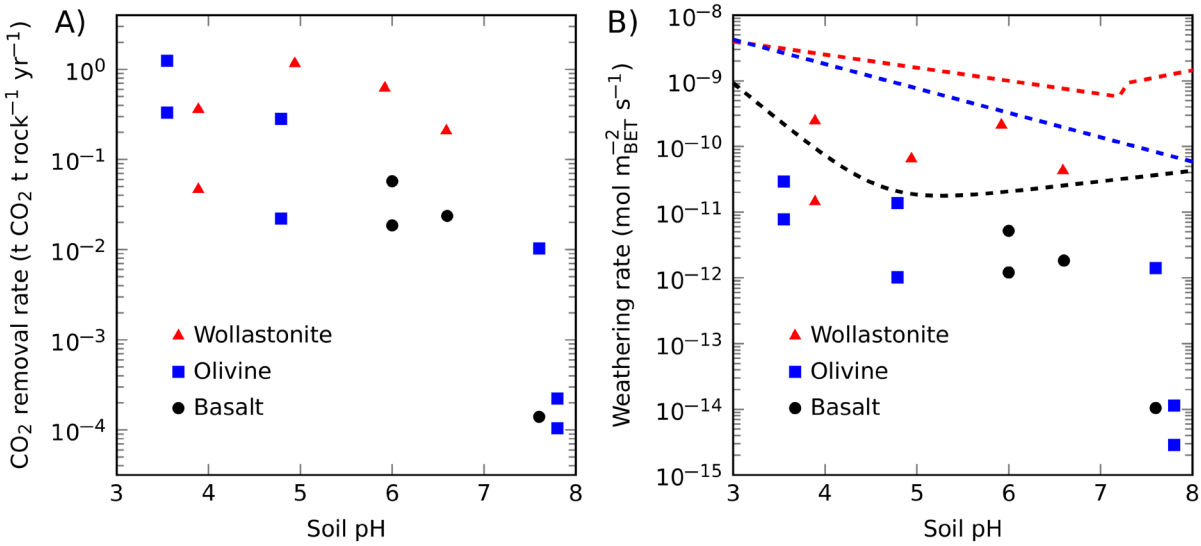


Fig. 2 Relationships Between CO_2 Removal Rate, Weathering Rate, and Soil pH. The CO_2 removal rate (A, normalized to the initial amounts the applied minerals) and weathering rate (B, normalized to the initial surface area of the applied minerals) exhibit a distinct dependency on pH, as evidenced by pilot- and field-scale observations of Enhanced Rock Weathering (ERW) (Table 3) across agricultural plots and natural catchments. These rates (represented as points) underscore the critical role of pH in the efficiency of ERW processes. A comparative analysis with the rate law derived from controlled lab-scale measurements (illustrated as dashed lines fitted using the kinetics rate law reported in Bandstra and Brantley 2008) highlights the potential variances in weathering kinetics under different environmental conditions

bands unless experiments are designed to separate carbonate and silicate contributions across both croplands and natural areas through pre-application carbonate-content screening, mineralogical characterization, and tracer-supported mass balance.

Isotopic and elemental tracers offer a path to attribution. Strontium isotopes can distinguish silicate from carbonate sources; radiocarbon constrains dissolved inorganic carbon ages; rare earth elements and immobile element ratios can fingerprint mineral dissolution and secondary precipitation pathways (Peters et al. 2004; Larkin et al. 2022; Kantola et al. 2023; Knapp et al. 2023; Reershemius et al. 2023). This coupling is especially important in natural ecosystems where agronomic yield signals are absent, yet the value of these tracers depends on coupling them to hydrologic measurements and to cation and alkalinity budgets. Without that coupling, tracer-based assignments do not resolve whether observed dissolution results in exported alkalinity or is offset by within-soil reprecipitation or by harvest removal of base cations. Consequently, Fig. 2 should not be interpreted as evidence that laboratory kinetics fail to predict field behavior, but rather as a prompt to jointly consider saturation indices, residence times, and tracer constraints in the same frame for both agricultural and natural sites so that field observations can be conditioned on the variables that actually control divergence from laboratory expectations.

4.2 Full lifecycle of ERW

Because field removal must be net of deployment emissions, lifecycle accounting is central to ERW evaluation. Figure 3 further reveals lifecycle CO_2 emissions, costs, and energy demands of ERW processes. Based on the data from Table 4 for each case study, we calculated the energy and emissions required to grind 1 tonne of rock. Grinding to finer particle sizes, such as 10 µm, results in emissions ranging from 0.18 to 163.55 kg of CO_2 and costs between $3.26 and $38.8, consuming approximately 0.002 to 1.2 GJ of energy (Fig. 3). This wide variation reflects differences in the hardness and mineral composition of the rock, regional electricity prices, and technological differences in grinding equipment across studies (Strefler et al. 2018; Kantzas et al. 2022; Li et al. 2024). Transport, particularly over

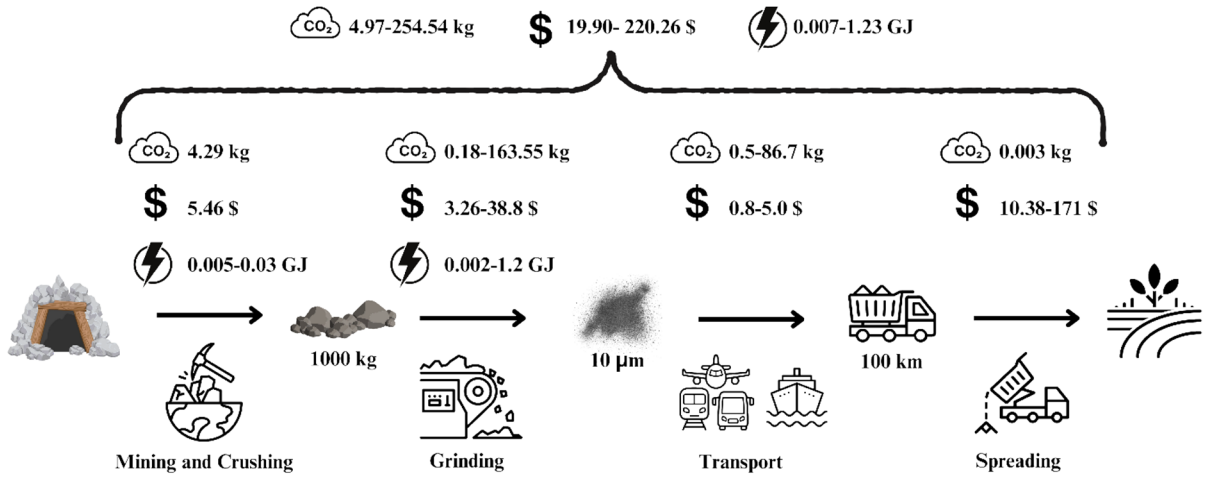


Fig. 3 Lifecycle CO_2 emissions, costs, and energy demands of ERW processes applicable to both agricultural and natural deployments. This figure provides a visual representation based on the data ranges from Table S1, which outlines the lifecycle CO_2 emissions, costs, and energy demands associated with key stages of the ERW process. These stages include mining and crushing, grinding, transport, and spreading. The example considers 1 tonne of rock being crushed to a particle size of 10 µm and transported over 100 km. The chart highlights the variation in energy input, emissions, and costs involved at each phase of the ERW process, supporting a comprehensive understanding of the environmental and economic impacts of ERW implementation

long distances like 100 km, contributes emissions between 0.5 and 86.7 kg of CO_2, with costs ranging from $0.8 to $5.0. Spreading of crushed rock adds further costs and emissions, ranging from $10.38 to $171 and producing approximately 0.003 kg of CO_2. This phase often rivals transport in terms of emissions and costs, especially when specialized equipment or large-scale deployment is required. In contrast, mining and crushing activities have a lower impact, emitting around 4.29 kg of CO_2 and costing $5.46, while consuming 0.005 to 0.03 GJ of energy. These insights suggest that improving the efficiency of grinding, optimizing logistics to reduce transport distances, and minimizing the energy required for spreading could significantly lower the overall carbon footprint and economic costs of ERW.

These lifecycle results shift the question from "can minerals dissolve" to "can they deliver net removal after energy and logistics are priced in." for both agricultural and natural deployments. Table 4 and Fig. 3 indicate that grinding dominates energy demand and emissions once particle sizes approach tens of micrometers, while mining and primary crushing are comparatively minor. Emissions and costs vary widely, which reflects differences in rock hardness and mineralogy, regional electricity prices and carbon intensity, and equipment performance across case studies (Strefler et al. 2018; Kantzas et al. 2022; Li et al. 2024).Transport is the second lever. Per kilometer emissions look small, but long hauls rapidly accumulate and can rival grinding when supply chains stretch beyond regional sources. Spreading can be either modest or material depending on method, with aerial application providing speed at the price of emissions and cost.

Two strategic implications follow. First, there is no universal "best" fineness across agricultural and natural settings. Because comminution energy rises steeply as target size falls (Renforth 2012; Strefler et al. 2018) and because field-effective surface area may decay over time (Sect. 2.3), net removal per unit energy has an interior optimum that depends on mineralogy, soil structure, and hydrology. Identifying that optimum requires pairing lifecycle inventories with measured alkalinity export for the same materials, sites, and grain size distributions rather than relying on laboratory kinetics alone. Second, logistics should be designed backward from candidate sites and sinks. Co-siting crushing and pre-grinding with existing quarries or industrial facilities, leveraging existing farm and forest road networks, prioritizing short supply chains, and aligning deployment with seasons

Table 4 Lifecycle CO_2 emissions, costs, and energy demands of ERW processes

Mining and crushing				
Study area	Lifecycle CO_2 emissions (kg CO_2 /t rock)	Cost ($/t rock)	Energy demand (GJ /t rock)	References
Global	N/A	N/A	0.01 and 0.03	(Goll et al. 2021)
UK	4.29	5.46	0.005	(Kantzas et al. 2022)

Grinding					
Study Area	Particle size (P80 values, μm)	Lifecycle CO_2 emissions of grinding (kg CO_2 /t rock)	Cost ($/t rock)	Energy demand (GJ /t rock)	References
Global	2	151.80–661.13	26.9–197	1.4–6.1	(Strefler et al. 2018)
	10	18.44–130.03	3.26–38.8	0.17–1.2	
	20	7.58–66.19	1.34–19.7	0.07–0.61	
	50	2.17–26.07	0.38–7.75	0.02–0.24	
U.S	10	0.18–71.36	3.63–11.84	0.002–0.66	(Li et al. 2024)
	50	0.06–23.10	1.18–3.83	0.001–0.21	
	100	0.03–11.67	0.59–1.94	0.000–0.11	
UK	10	89.4–163.55	16.01–19.01	0.005	(Kantzas et al. 2022)
	100	24.09–44.03	8.89–11.51	0.001	

Transport				
Study area	Lifecycle CO_2 emissions of transportation (kg CO_2/t/km)	Cost ($/t rock/ km)	Maximum transportation distances (km) for achieving a net carbon reduction	References
Global	0.005 to 0.867	0.0086–0.009	450 for air transport (based on 280 kg CO_2 per tonne of basalt) 1,000 in remote regions (e.g., Amazonia or Siberia, with higher CDR potential)	(Goll et al. 2021)
Global	N/A	0.05	Transport for temperate and warm regions, covering 300 km, may be conducted by road, rail, or ship	(Strefler et al. 2018)
UK	N/A	0.008 (1000 km)	For road freight, the transport technology transition from the E3ME[a] for electric vehicles was adopted	(Kantzas et al. 2022)
U.S. Midwest croplands	0.02 to 0.5	0.0185	552–1109 (combined barge and truck) with a CDR yield of 0.35 tCO_2/t Basalt	(Zhang et al. 2023a)
Brazil	0.08 to 0.26	N/A	300	(Jerden et al. 2024)

Spreading				
Study Area	Lifecycle CO_2 emissions of spreading (kg CO_2 /t rock)	Cost ($/t rock)	Description	Reference
Global	N/A	88–171	Helicopters or small fixed-wing aircraft equipped with agricultural spreaders that can carry up to 4 t over a range of ~1,000 km	(Goll et al. 2021)
UK	0.003	10.38	Spreading costs were assigned equally to equipment, fuel/electricity and wages	(Kantzas et al. 2022)

a: Global E3 (Energy-Environment-Economy) macro-econometric model.

that maximize hydrologic contact (e.g., wet seasons in croplands and high-flow periods in forested headwaters) can reduce both costs and emissions. Mine tailings and industrial by-products are especially relevant in this context because they may already occur at favorable grain sizes as part of conventional extraction or processing, thereby reducing the additional comminution energy assigned to ERW. However, this advantage is only real when the material has suitable mineralogy, low contaminant risk, consistent particle-size distribution, and a defensible allocation of pre-existing processing emissions (Moosdorf et al. 2014; Goll et al. 2021; Kantzas et al. 2022; Li et al. 2024). These choices are material to viability since the range in grinding and transport burdens shown in Fig. 3 is comparable to, and sometimes larger than, the removal credited in field studies.

Finally, boundary conditions in lifecycle assessments must be explicit. Reported emissions should be tied to marginal rather than average grid intensities where relevant, capital amortization should be transparent, and coproduct credits, if any, should be justified rather than assumed, including explicit treatment of biomass harvest exports and land-management operations where applicable in natural systems. Without these conventions, cross-study comparisons collapse into accounting artifacts even when underlying engineering is sound (Strefler et al. 2018; Kantzas et al. 2022; Li et al. 2024).

4.3 Economic considerations

Economic assessment builds directly on lifecycle accounting because costs, emissions, and creditability are linked. Economic viability of ERW depends on the joint distribution of costs for comminution, transport, and spreading, on the durability and measurability of removal, and on any agricultural co-benefits that persist once nutrient budgets and harvest exports are accounted for (Fig. 4). In agricultural systems, co-benefits may include yield gains; in natural systems, benefits center on acid neutralization, base-cation



Fig. 4 Enhanced Rock Weathering Process and Impact Analysis (reflecting both agricultural and natural ecosystems). This infographic summarizes major ERW stages, including mining, grinding, transport, spreading, mineral dissolution, CO_2 uptake, and potential storage as dissolved inorganic carbon or secondary carbonates. It distinguishes agricultural deployments, where yield and soil-fertility co-benefits may partly offset costs, from natural systems, where acid neutralization, base-cation replenishment, water-quality gains, and biodiversity safeguards are more relevant. The figure also highlights major cost and emission sources, including comminution, transport, and spreading, and separates agriculture-specific co-benefits from non-agricultural ecosystem services to avoid double counting in MRV. Land-use-specific risks, including trace metals, sediment loss, habitat disturbance, and monitoring requirements, are shown as safeguards that must be evaluated alongside carbon-removal claims

replenishment, and water-quality improvements rather than crop yields. The mesocosm and field examples indicate that basalt applications can raise yields in some contexts, and multi-year trials in major cropping systems report double-digit increases in maize and soybean production, which can partially offset deployment costs (Kelland et al. 2020; Beerling et al. 2024). Yet yield gains are not fungible with removal credits unless the same experiments document alkalinity export and cation budgets. Otherwise, revenue stacking risks double counting benefits that draw on the same underlying processes. Recent system-scale analyses for the US Corn Belt by Beerling et al. (2025a) refine these economic projections, estimating costs between $95 and $170 per tonne of CO_2, depending heavily on transport distances. Crucially, they identify that economic viability is fragile; it relies not just on rock costs, but on the rigorous verification of removal. If MRV protocols cannot distinguish between fertilizer-induced acidification and rock-induced weathering, the economic value of the carbon credit collapses (Clarkson et al. 2024).

Regional heterogeneity is the rule. Energy prices, labor, equipment access, and suitable lithologies vary across settings, which produces broad cost bands in practice and widens the uncertainty on scale-up trajectories (Doetterl et al. 2018; Strefler et al. 2018; Smith et al. 2019), and land-use constraints (e.g., access and permitting on public lands) can further shape feasibility in natural areas. In croplands, existing spreading equipment, farm records, and yield data can reduce marginal deployment and monitoring costs. In natural ecosystems, by contrast, access limits, lower tolerance for soil disturbance, absence of crop-revenue offsets, and greater need for hydrologic and ecological baselines make ERW more suitable for carefully designed pilots than for immediate broad deployment. Urban and constructed-soil settings may provide useful controlled test beds, but their limited area, engineered drainage, and infrastructure constraints make them complements rather than substitutes for agricultural or watershed-scale deployment (Haque et al. 2021; Jorat et al. 2022; Kantzas et al. 2022; Levy et al. 2024). Reuse of mining and industrial by-products can reduce feedstock costs and, in some cases, lower environmental liabilities, but the LCA benefit depends on whether these materials are genuinely pre-ground coproducts rather than materials requiring ERW-specific comminution. Quality control on trace metals and particle size distributions becomes essential for both agronomic acceptance and MRV integrity (Stavins 1999; Moosdorf et al. 2014; Psarras et al. 2020; Goll et al. 2021), with screening tailored to sensitive natural habitats. A defensible economic case therefore links site-specific lifecycle costs to verifiable removal and to clearly separated agronomic benefits or non-agricultural co-benefits, and it expresses uncertainty through sensitivity to power prices, haul distances, and grain size targets rather than through point estimates.

4.4 Policy considerations

Policy considerations follow from this economic uncertainty because credit value depends on credible rules for measurement, safety, and accountability. Policy determines whether ERW remains a collection of local experiments or evolves into a managed carbon removal option with standards that guard against over-crediting and environmental risk in both agricultural and natural landscapes. Jurisdictions with mature environmental policy frameworks can integrate ERW into existing instruments if MRV protocols are strong enough. Policy must move beyond ‘pay-for-practice’ (paying for spreading rock) to ‘pay-for-performance’ (paying for verified alkalinity flux). This requires standardized protocols that explicitly account for non-carbonate sources of alkalinity and deduct potential organic carbon losses triggered by priming effects (Wolf et al. 2024; Beerling et al. 2025a). However, performance-based systems also need to remain accessible to farmers; otherwise, high MRV costs and carbon-market gatekeeping may slow adoption despite ERW’s agronomic potential (Power et al. 2026). In settings where resources and regulatory capacity are limited, policy emphasis may rationally shift toward soil enhancement and food security, particularly where croplands are extensive and P limitation is widespread, while in natural landscapes the emphasis may be on acid mitigation and watershed health. However, this strategy still requires guardrails and support for measurement to avoid trading near-term agronomic gains for long-term environmental costs (Renforth et al. 2009; Beerling et al. 2020; Holzer et al. 2023; Mir et al. 2023; Schenuit et al. 2024).

Because environmental benefits and hydrologic impacts cross borders and many natural headwaters

are transboundary, international collaboration has a direct bearing on ERW standards. Harmonized definitions of removal, shared reference methods for alkalinity and tracer measurements, and interoperable data formats would improve comparability and reduce the risk of cherry-picking favorable accounting. Equity considerations also matter. Policies should ensure that the communities who host mining, processing, and spreading retain agency and gain tangible benefits, and that public health and education are integrated with deployment where appropriate (Lawford-Smith and Currie 2017; Cox et al. 2022; Vandeginste et al. 2024), including for communities adjacent to natural areas slated for ERW pilots. Figure 4 is thus most useful when read as a policy design checklist. It explicitly distinguishes pH adjustment from alkalinity export, offers parallel risk indicators for agricultural and natural deployments (e.g., trace metals, sediment loss, habitat disturbance), and promotes regional planning that aligns feedstock supply with demand while keeping haul distances manageable. Effective frameworks will remain flexible, but they should be anchored in mass-balance-based MRV and in protections that sustain soil and water quality alongside carbon accounting and adapted to land-management regimes outside agriculture.

5 Modeling perspectives in ERW

Table 5 summarizes current ERW modeling approaches by the range of processes represented ("Model types") and by their spatial and temporal scales, which in turn determine the types of problems they are best suited to address and their respective strengths and limitations. Because ERW involves coupled reactions, transport, biology, logistics, and governance, no single model tier can resolve all sources of uncertainty.

Reactive transport models provide the most explicit representation of coupled chemistry and transport at pore to plot scales. They can track dissolution, precipitation, diffusion, and evolving saturation states, which makes them indispensable for evaluating hypotheses about passivation, silica feedbacks, and residence time effects on effective rates. Yet they often operate under boundary conditions that are easier to impose in a laboratory than to verify in a field setting, and they rarely include dynamic biological forcings such as root uptake and microbial production of ligands at relevant temporal resolutions (Pandey and Rajaram 2016; Deng et al. 2023). Reliance on theoretical rate constants and BET-derived surface areas in reactive transport models may have led to systematic overestimation of dissolution rates (Sect. 2.3; Power et al. 2025). Computational burden then limits ensemble sizes and the exploration of parameter non-uniqueness. For decision support, the clearest contribution of this tier is to produce calibrated, transferable relationships between effective rate, saturation index, and residence time that downstream models can use, rather than to deliver single-scenario forecasts that are difficult to validate. Taken together, these pore- to plot-scale constraints define effective rate–saturation–residence-time relationships that directly feed the next tier of soil and ecosystem models.

Soil and ecosystem models extend the domain to seasons and years and incorporate plant–water–soil interactions with explicit dynamics. They can represent mixing depths, infiltration and drainage, cation exchange, and biological enhancement of dissolution while tracking management interventions such as tillage and harvest. Their geochemical cores are necessarily simplified, and mineralogies are often reduced to a few classes, which introduces structural uncertainty that is easily hidden by site-specific calibration (Taylor et al. 2016; Cipolla et al. 2021, 2022). This tier is most informative when it reports mass-balanced alkalinity export, cation budgets that include plant uptake and removal at harvest, and distributions of residence times. Without that suite of outputs, agreements with pH or yield trajectories risk being diagnostic of agronomy rather than of carbon removal. Those mass-balanced exports and residence-time distributions then provide boundary conditions and priors for Earth system models.

Earth system models link land to river networks and the ocean, where carbon permanence is ultimately established. These models can explore teleconnections and policy-relevant scales, including the response of marine carbonate chemistry to sustained alkalinity inputs. Their key strength lies in the representation of feedbacks between land, ocean, and atmosphere over decades to centuries. However, a common weakness is the tendency to assume rapid or instantaneous export of alkalinity from soils and rivers, which likely overstates sequestration when

Table 5 Hierarchical classification of ERW modeling approaches, key features, and limitations

Model types	Representative models	Spatial/temporal scale	Key strengths	Key limitations	references
Reactive-transport	PFLOTRAN, TOUGHREACT, CrunchFlow	cm–m; hours–months	High-resolution geochemistry; detailed kinetics; explicit diffusion & precipitation modeling	High computational cost; limited biological processes; challenging to scale up or validate in field	(Pandey and Rajaram 2016) (Deng et al. 2023)
Soil and ecosystem biogeochemical	Sheffield Model, Soil Model for Enhanced Weathering	m–km; seasons–decades	Integrates soil hydrology, plant interactions, and biological weathering	Simplified mineralogy; uncertain parametrization; often site-specific calibration	(Taylor et al. 2016)(Cipolla et al. 2021, 2022) (Bertagni et al. 2025)
Earth system models	GENIE, cGENIE (w/ weathering modules)	Global; decades–centuries	Captures feedback with atmosphere–ocean; can simulate large-scale deployment and environmental impact	Oversimplified terrestrial processes; instantaneous bicarbonate export assumption may overestimate sequestration	(Maffre et al. 2018)(Vakilifard et al. 2021)(Vandeginste et al. 2024)(Val Martin et al. 2023)(Taylor et al. 2016)
Integrated assessment and optimization models	WITCH, GCAM, MILP logistics frameworks, c-RISE	Regional–global; multi-decadal	Incorporates costs, land competition, grid-carbon intensity, policy scenarios; aligns with CDR deployment needs	Relies on coarse assumptions from upstream models; rarely includes uncertainty bounds or mechanistic constraints	(Goddéris et al. 2006) (Zhang et al. 2023b) (Zhang et al. 2023a) (Morrow et al. 2023) (Kantzas et al. 2022)

real flow paths include storage, re-precipitation, and biogeochemical attenuation (Maffre et al. 2018; Vakilifard et al. 2021; Vandeginste et al. 2024). In other words, this 'perfect transport' assumption likely overstates sequestration. Zhang et al. (2025) recently introduced a framework for modeling riverine carbon loss, demonstrating that a significant fraction of soil-derived bicarbonate can re-equilibrate and degas as CO_2 during transit through river networks. Therefore, Earth system models that lack explicit riverine evasion modules risk overestimating the durability of ocean storage. To be credible, global projections must couple terrestrial weathering rates with hydrodynamic routing that accounts for this downstream leakage. The most constructive way to use this tier is to propagate calibrated fluxes and uncertainties from the terrestrial models, to include realistic routing and transformation in rivers, and to make the lag between application and ocean uptake an explicit, reportable quantity. The resulting long-horizon alkalinity and carbon trajectories inform integrated assessment and optimization models.

Integrated assessment and optimization models translate biogeophysical potential into deployment pathways. They evaluate source–sink logistics, power mix, land competition, and policy scenarios and are therefore essential for comparing ERW with other options within constrained portfolios (Goddéris et al. 2006; Kantzas et al. 2022; Morrow et al. 2023; Zhang et al. 2023a, b). Recent advances have moved beyond coarse global averages. For instance, Beerling et al. (2025a) utilized high-resolution datasets for US agriculture to map the spatial heterogeneity of deployment costs and removal potential. By integrating granular soil pH buffering capacity and proximity to rail networks, such models filter out theoretical 'hotspots' that are economically or agronomically unviable, providing a more conservative but actionable roadmap for deployment. However, they sit at the end of the modeling chain and inherit every upstream simplification. When average removal rates, constant costs, and thin uncertainty bounds are supplied, large-scale projections reflect accounting choices more than physical limits. Their added value grows when they accept ensembles of removal and cost trajectories conditioned on grain size, residence time distributions, and lifecycle burdens, and when they return the implied infrastructure and power demands back to the process tiers for feasibility checks. In short, each tier conditions the next, and feedbacks back to process-level feasibility need to be explicit.

To connect these modeling tiers with the knowledge gaps discussed later, we summarize cross-tier implications and practical gaps below and then point forward to broader recommendations.

From the model development perspective, reactive transport models should consider the modeling of "effective" surface areas that may be lower than BET areas due to pore inaccessibility, passivation, and changing grain size distributions over time. Soil and ecosystem models should commit to validation on measured alkalinity exports with matched cation budgets. Hydrological residence time is a key factor in the field, and both reactive-transport and soil-and-ecosystem models can benefit from more realistic meteorological inputs and sensitivity analyses that separate hydrological transport limitation from dissolution rate limitation. Earth system models should integrate explicit riverine evasion modules to account for CO_2 degassing during transit and separate alkalinity injection from long-term oceanic storage. Integrated assessment models should utilize high-resolution spatial datasets to filter out theoretical 'hotspots' that are economically unviable and commit to reporting the marginal power and logistics requirements implied by any scale-up.

For validating modeled CDR rates, field experiments and watershed-scale studies are invaluable for identifying discrepancies between theoretical and field-scale carbon removal, as well as hydrological transport bottlenecks (e.g. (Peters et al. 2004; Bertagni et al. 2025)). Validation of biological interactions such as soil organic carbon response and nitrogen oxides emissions can reveal co-benefits and trade-offs in addition to the direct, dissolution-associated removal (Val Martin et al. 2023). The terrestrial component of ERW must be further coupled with validated models of riverine carbon loss, travel time to the ocean, and ocean storage efficiency to avoid projecting 'phantom' weathering (Vakilifard et al. 2021; Zhang et al. 2025). A coherent validation program should align metrics across tiers so that what is verified at one scale becomes a conditioned prior at the next.

Information should also move up the spatiotemporal hierarchy in a form that preserves the underlying physics–for example, by providing distributions or transferable relationships between effective

weathering rates and saturation state rather than single mean values. There should also be bidirectional feedbacks where implied infrastructure and power demands from assessment models are returned to process tiers for feasibility checks. This ensures that the hierarchy functions as a set of coupled constraints rather than disconnected tools.

Finally, claims about the carbon removal potential of ERW should always be presented together with the modeling assumptions on which they are based. Statements about deployment potential or system costs that cannot be traced to the underlying assumptions of the dissolution model (e.g., rate laws, BET versus effective surface area, and the scope of geochemical, hydrological, and biological processes represented), as well as assumptions regarding riverine and oceanic carbon transfer efficiency and lifecycle emissions, are unlikely to withstand cross-scale or cross-model scrutiny. Conversely, when those choices are explicit and validated at each tier, the hierarchy can deliver predictions that are both decision relevant and scientifically defensible.

6 Comparative analysis and risks of ERW

6.1 Comparison of ERW with other CDR techniques

Comparative assessments of carbon removal options are most informative when system boundaries, baselines, and monitoring requirements are aligned. Under this framing, Fig. 5 functions as a benchmark for sensitivity rather than a definitive ranking. The salient question is which assumptions need to hold

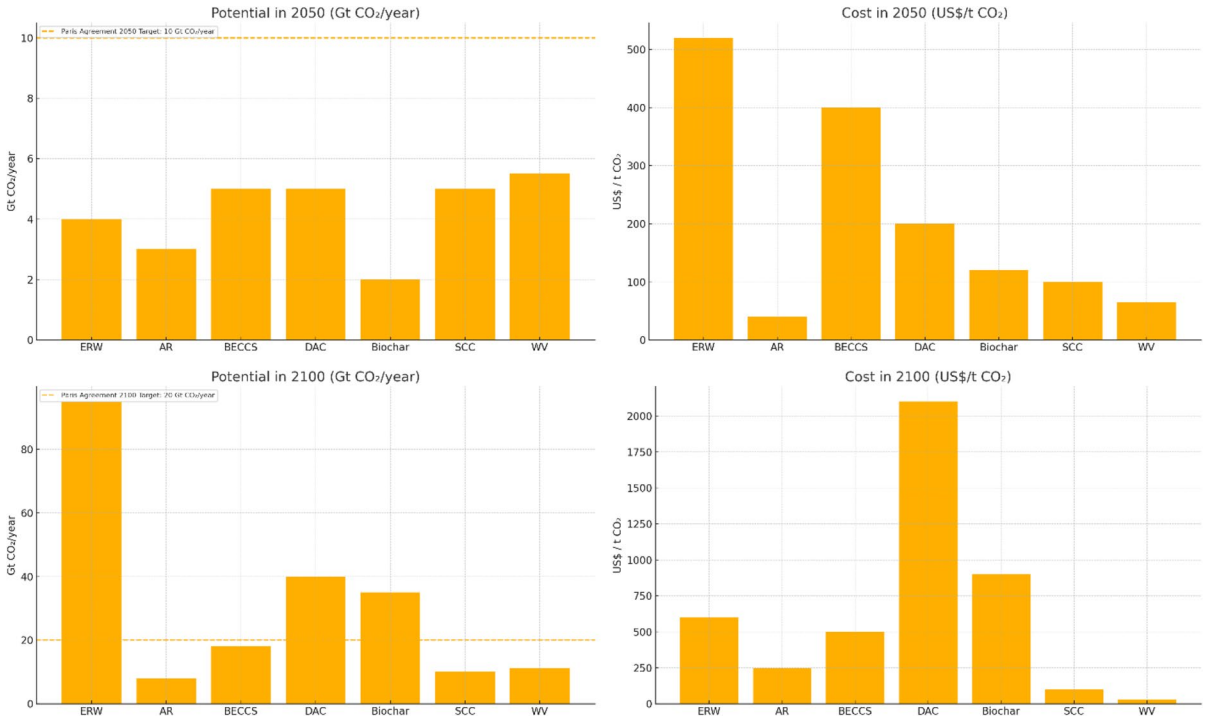


Fig. 5 Ranges of potential and cost for carbon dioxide removal methods by 2050 and 2100. This figure is based on the data presented in Table 6. It compares the potential (in Gt CO_2/year) and cost (in US$/t CO_2) of several CDR technologies for 2050 and 2100. The CDR methods include Enhanced Rock Weathering (ERW), Afforestation and Reforestation (AR), Bioenergy with Carbon Capture and Storage (BECCS), Direct Air Capture (DAC), Biochar, Soil Carbon Sequestration (SCC), and Wood Vaulting (WV). The orange boxes represent the projected range between the minimum and maximum potential and cost values for each method. The dashed orange lines represent the targets of 10 Gt CO_2/year by 2050 and 20 Gt CO_2/year by 2100, as estimated by the National Academy of Sciences (National Academies of Sciences and Medicine 2019), indicating the scale of CDR required annually to meet the Paris Agreement's goals. Projected cost ranges may rise in 2100 due to scale-related factors like land and resource constraints, even as technology matures

Table 6 Comparison of ERW with other CDR Techniques

CDR approach	Time	Potential (Gt CO_2/year)	Cost (US$/ t CO_2)	References
ERW	**2050**	**0.5–2**	**160–180**	(Beerling et al. 2020)
		2–4	**50–200**	(Fuss et al. 2018)
		0.2–2.5	**79–513**	(Goll et al. 2021)
	2100	**N/A**	**~240**	(Prado and Mac Dowell 2023)
		5	**~600**	(Strefler et al. 2021)
		< 1–95	**24–578**	(Pathak et al. 2022)
AR	**2050**	**0.5–3.6**	**5–50**	(Fuss et al. 2018)
	2100	**0.5–7**	**N/A**	(Brack and King 2021)
		N/A	**~28–50**	(Prado and Mac Dowell 2023)
		1.1–3.3	**65–108**	(Smith et al. 2016)
		0.5–10	**0–240**	(Pathak et al. 2022)
BECCS	**Current**	**N/A**	**125–285**	(Fahs et al. 2023)
	2050	**0.5–5**	**100–200**	(Fuss et al. 2018)
	2100	**3.3**	**132**	(Smith et al. 2016)
		~9	**500**	(Strefler et al. 2021)
		0.5–11	**15–400**	(Pathak et al. 2022)
		1–20+	**N/A**	(Brack and King 2021)
		N/A	**~190**	(Prado and Mac Dowell 2023)
DAC	**Current**	**N/A**	**225–600**	(Fahs et al. 2023)
	2050	**0.5–5**	**100–300**	(Fuss et al. 2018)
		N/A	**65–145**	(Fahs et al. 2023)
	2100	**N/A**	**1,600–2,080**	(Smith et al. 2016)
		1–20+	**N/A**	(Brack and King 2021)
		N/A	**~620–830**	(Prado and Mac Dowell 2023)
		1.5	**N/A**	(Realmonte et al. 2019)
		5–40	**84–386**	(Pathak et al. 2022)
		3.8	**~630**	(Strefler et al. 2021)
Biochar	**Current**	**N/A**	**90–120**	(Fahs et al. 2023)
	2050	**0.5–2**	**30–120**	(Fuss et al. 2018)
	2100	**1–35**	**N/A**	(Brack and King 2021)
		N/A	**200–800**	(Prado and Mac Dowell 2023)
		0.3–6.6	**10–345**	(Pathak et al. 2022)
SCC	**2050**	**2–5**	**0–100**	(Fuss et al. 2018)
	2100	**0.5–11**	**N/A**	(Brack and King 2021)
		0.6–9.3	**-45–100**	(Pathak et al. 2022)
Wood vaulting	**2050**	**1–10**	**30–100**	(Zeng and Hausmann 2022; Zeng et al. 2024)
	2100	**10.1–12.4**	**10–50**	(Luo et al. 2025)

CDR potentials are net values (subtracting lifecycle emissions) unless otherwise noted in the source; refer to individual references for detailed accounting methodologies

for each option to deliver verifiable removal at stated scales and costs, and how robust those assumptions appear in real landscapes and energy systems.

ERW couples potential carbon removal with agronomic services. Reported mid-century ranges span approximately 0.2 to 4 Gt CO_2 per year at costs of about 50 to 513 US dollars per tonne, while late-century scenarios extend to values up to 95 Gt CO_2 per year with costs from roughly 24 to 600 US dollars per tonne (Strefler et al. 2021; Pathak et al. 2022). These figures are useful for exploring parameter space, yet they embed choices about grain size targets,

passivation, residence times, and supply-chain emissions. Field evidence indicates that such choices materially affect outcomes, so ERW potentials are best treated as conditional on-site chemistry and hydrology, as well as on MRV focused on alkalinity export and cation budgets (Beerling et al. 2020; Goll et al. 2021; Zhang et al. 2023a).

Direct Air Capture offers location flexibility and comparatively straightforward accounting of captured CO_2, although energy intensity typically governs cost. Mid-century projections of about 0.5 to 5 Gt CO_2 per year at 100 to 300 US dollars per tonne are consistent with this trade-off; late-century estimates near 40 Gt CO_2 per year often carry wide cost bands that reflect constraints in low-carbon power and infrastructure (Ruthven 2014; Zeman 2014; Smith 2016; Fuss et al. 2018). Afforestation and reforestation provide biodiversity and hydrologic co-benefits but store carbon over longer horizons and are sensitive to land and water competition. Reported potentials of roughly 0.5 to 3.6 Gt CO_2 per year by 2050 and up to about 10 Gt CO_2 per year by 2100 occur at relatively low average costs, provided sustained stewardship and appropriate species–site matching (Farley et al. 2005; Anderson et al. 2014; Lautenbach et al. 2017; Fuss et al. 2018; Pathak et al. 2022).

Bioenergy with Carbon Capture and Storage (BECCS) integrates energy production and storage but inherits land and nutrient constraints associated with biomass, alongside capture and transport requirements. Mid-century ranges of approximately 0.5 to 5 Gt CO_2 per year at 100 to 400 US dollars per tonne and late-century values up to about 20 Gt CO_2 per year are contingent on sustainable feedstock supply and competing land uses (Fuss et al. 2018; Strefler et al. 2021). Biofuels and soil carbon sequestration occupy different positions in the portfolio. Net outcomes for biofuels remain sensitive to crop choice, management, and coproduct accounting, which complicates generalization and may limit scalability where arable land is constrained (Holzman 2008; Tittmann et al. 2010; Kochsiek and Knops 2012; Searchinger et al. 2015; Johansson et al. 2020). Soil carbon sequestration and biochar often appear lower cost and nearer term, yet both face monitoring and permanence challenges and show pronounced spatial variability. Literature ranges include several gigatonnes per year by 2050 and larger values by 2100 with wide cost bands (Lal 2004; Woolf et al. 2010; Smith 2016; Fuss et al. 2018; Brack and King 2021; Pathak et al. 2022). Wood vaulting (WV) stores woody biomass in engineered, oxygen-limited and dry vaults to slow decomposition. Reported mid-century ranges are roughly 1–10 Gt CO_2 yr^{-1} at US$ 30–100 t^{-1} CO_2, while long-run scenarios average 10.1–12.4 Gt CO_2 yr^{-1} at US$ 10–50 t^{-1} CO_2 under optimized logistics (Zeng and Hausmann 2022; Zeng et al. 2024; Luo et al. 2025).

Taken together, ERW appears most compelling where soils are acidic, phosphorus limited, and hydrologically suited to move alkalinity to streams without substantial loss or reprecipitation. In such settings, soil services can offset part of the cost while MRV focuses on export rather than proxy variables. Where baseline carbonate inputs are common or residence times are short, observed gains are likely to be smaller unless management is adjusted. WV may complement these portfolios where woody residues are abundant and storage sites are proximate, offering comparatively low energy dependence but high siting and MRV dependence. Given National Academies targets near 10 Gt CO_2 per year by 2050 and about 20 Gt CO_2 per year by 2100, portfolio design benefits from combining options with less-correlated uncertainties and credible monitoring pathways at scale (Haszeldine 2009; Montzka et al. 2011; Hoegh-Guldberg et al. 2019; IPCC 2023).

6.2 Adverse effects of ERW

Large-scale deployment invites attention to environmental health and social acceptance alongside carbon accounting. Trace metal mobilization is not merely a side effect but a potential threat to ecosystem function. Ultrabasic and some mafic feedstocks contain nickel and chromium that can be released during weathering, and field and mesocosm studies indicate that repeated or heavy applications may raise soil concentrations toward regulatory thresholds within decades at higher application rates (Hartmann et al. 2013; Haque et al. 2020b; Dupla et al. 2023). Recent risk assessments by Levy et al. (2024) and Dupla et al. (2023) demonstrate that under high application rates, Ni and Cr accumulation can reach thresholds that are toxic to earthworms and soil invertebrates, effectively reversing the ‘synergies’ discussed in Sect. 3. Furthermore, trophic transfer of these metals into crops poses a

food safety risk that could disqualify ERW products from food supply chains. Material choice therefore embeds a risk–benefit calculus. Ultramafic sources rich in olivine offer higher intrinsic reactivity but also carry Ni, Co, and Cr that can mobilize under weathering and accumulate in plants and soils, which raises monitoring and permitting hurdles (Alexander 2007; ten Berge et al. 2012a, b; Amann et al. 2020). Basalts are more widely available and agronomically practical, often delivering nutrients along with alkalinity, although unit-mass CO_2 removal is typically lower than for more reactive ultramafic materials. Where local supply chains and geochemical screening support its use, Ca-silicate feedstock such as wollastonite can provide an alternative with high weathering potential and lower Ni-leaching concern than olivine in comparative soil-column studies (Middlemost 1975; Strefler et al. 2018; Goll et al. 2021; Lewis et al. 2021; Beerling et al. 2024). Consequently, feedstock selection must be governed by strict geochemical safety limits, prioritizing basalts with lower heavy metal indices over more reactive but riskier ultramafic rocks. Metals often sorb to iron oxides, which can limit near-term leaching while establishing reactive pools that are susceptible to remobilization under reducing or highly acidic microsites. Observations in soils and waters implicate adsorption and subsequent transport pathways for copper, lead, chromium, manganese, and nickel (Okereafor et al. 2020; Saikia et al. 2022; Dupla et al. 2023). Feedstock screening, pre-application leaching tests, and post-application monitoring are therefore prudent program elements. Trace-metal thresholds should be reported by matrix rather than as a single value: total or extractable Ni and Cr in solid soil indicate accumulation risk, dissolved Ni and Cr in porewater indicate near-term bioavailability and leaching risk, and crop or edible-tissue concentrations indicate food-safety relevance. These thresholds should be compared against local soil-quality standards, ecotoxicological endpoints for soil biota, and crop-specific food-safety limits where applicable (Dupla et al. 2023; Levy et al. 2024). Where thresholds are approached or exceeded, the first response should be to reduce application rates, change feedstock, pause deployment, or redesign monitoring before relying on remediation. Where risk is elevated by legacy contamination or sensitive receptors, phytoremediation with hyperaccumulator species has been proposed, ideally coupled with careful biomass handling to avoid secondary releases (Suhrhoff 2022).

Soil chemistry responses point to additional trade-offs. ERW can raise pH and improve nutrient availability in acidic soils, yet in alkaline settings further pH elevation can reduce the availability of key nutrients and depress yields (Farhangi-Abriz et al. 2021). Short-term pulses in soil respiration and trace gas fluxes, including CO_2 and CH_4 in organic-rich soils, may erode near-term gains and complicate attribution if alkalinity export is not measured concurrently (Beerling et al. 2018; Dupla et al. 2024). Agronomic deployment is therefore better conditioned on baseline chemistry, buffering capacity, and crop removal of base cations, with adaptive management that responds to observed rather than assumed benefits.

Human health risks constitute a second critical barrier. Worker and community health risks arise mainly from dust during grinding, handling, and spreading. Fine particulates pose respiratory hazards, and the possibility of asbestos in some serpentine-bearing ultramafic rocks warrants screening and careful handling to meet occupational and public health standards (Dostert et al. 2008; Vandeginste et al. 2024). Vandeginste et al. (2024) highlight that the fine particulate matter (PM2.5 and PM10) generated during the comminution of silicate rocks poses chronic respiratory hazards comparable to mining dust exposure. Moreover, the potential presence of asbestiform minerals in specific serpentine-bearing olivine sources necessitates rigorous mineralogical screening (Oppon et al. 2024b). Practical mitigation includes feedstock characterization for fibrous phases, dust suppression, enclosed conveyance where feasible, appropriate protective equipment, and near-field air-quality monitoring during application.

Landscape and socioeconomic risks accompany efforts that aim for very large scales. Application across extensive areas can interact with other land uses, fragment habitats, and increase logistics burdens. Long-term costs for monitoring, risk management, and potential remediation remain uncertain and could challenge claims of low cost if carbon removal efficiencies vary across ecological and climatic gradients (Beerling et al. 2020; Dupla et al. 2023; Kantola et al. 2023; Vandeginste et al. 2024). Programs

that align feedstock sourcing with existing quarries or industrial by-products, target soils where chemistry and hydrology favor alkalinity export, and embed monitoring from the outset tend to be more resilient than programs guided primarily by theoretical removal estimates.

Moving forward, risks of ERW can be mitigated by a set of integrated safety practices. On feedstock safety, application rates should be governed by Ni and Cr thresholds referenced to local soil backgrounds and the sensitivity of resident biota, and source material should be screened for fibrous mineral phases and pre-leached where fines pose elevated risk. For MRV, the cation mass balance should be closed and alkalinity export reported so that treatment effects can be distinguished from legacy inputs. Site selection and application timing should account for local hydrology: slopes and soils that maximize water–mineral contact while minimizing erosion should be prioritized, and spreading should be aligned with wet-season infiltration periods. Wetlands and high-DOC hotspots that favor CH_4 production should be avoided. On the supply chain side, lifecycle emissions can be reduced by co-site crushing with quarries/industrial plants, using low-carbon electricity, and targeting the interior optimum of "net removal per unit energy" rather than minimum particle size. Finally, occupational health and community acceptance require clear dust-control plans, appropriate buffer distances, and accessible grievance mechanisms, along with tangible local benefits such as employment opportunities, improved soil services, and open access to monitoring data. These measures shift ERW from literature synthesis to deployable practice by linking site screening, MRV, and LCA to explicit guardrails and adaptive management.

7 Knowledge gaps

7.1 Geological inconsistencies and unknowns

Despite rapid progress, several geological and spatial uncertainties continue to limit ERW deployment planning. ERW performance varies with geology and geography, yet the spatial resolution and completeness of mineralogical maps remain uneven across regions, which limits siting and scaling analyses (Strefler et al. 2018; Calabrese et al. 2022). Mineral composition is recognized as a first-order control, although global identification of favorable lithologies is still incomplete. Soil properties then modulate outcomes through multiple pathways. Routine descriptors such as texture, pH, organic matter, and bulk cation exchange capacity capture part of the variance, while macroporosity, mineralogical assemblages, pore water chemistry, and cation exchange capacities disaggregated by ion and organic matter dependence often govern contact times and reaction trajectories yet are less frequently measured or mapped (Haque et al. 2019; Abdalqadir et al. 2024).

Hydrogeochemical context further conditions weathering rates. Position in the watershed, interactions between soils and shallow groundwater, bedrock weathering, and groundwater composition can shift saturation states and transport pathways, thereby altering apparent rates and the fate of alkalinity (Velbel and Price 2007; Bailey et al. 2014; de Oliveira Garcia et al. 2020; Zhang et al. 2022). These interactions are not fully represented in present siting frameworks. Ecological feedbacks add another layer of uncertainty. Microbial communities, plant health, and water quality can respond to ERW in ways that either reinforce or counteract intended geochemical effects, and these responses are likely to vary with setting (Edwards et al. 2017). A priority is to link mineral resource mapping with soil functional traits and hydrogeologic domains at resolutions relevant to deployment. Doing so would allow models to move from lithology-only screening to coupled assessments that include contact times, saturation regimes, and ecological risk indicators.

7.2 Data gaps and experimental shortcomings

These geological uncertainties are compounded by fragmented observations and inconsistent experimental designs. Progress is constrained by fragmented observations and heterogeneous methods. Long-term field studies remain sparse, even though they are well suited to evaluate persistence of alkalinity export, soil health trajectories, and ecological responses (Edwards et al. 2017). Short experiments provide mechanistic insight but often lack the temporal coverage needed to separate transient pH adjustment from durable export. Most legacy datasets report changes in soil solid phases or pH (state variables). There is a scarcity of data that closes the budget between total dissolved

alkalinity and exported alkalinity from the soil, creating a 'verification gap' that prevents the validation of the reactive-transport and soil-and-ecosystem models in Table 5 (Clarkson et al. 2024). Cross-scale data are also limited. Molecular-scale processes like secondary precipitation and surface passivation are rarely measured, and plot-scale findings are not consistently nested within catchment monitoring (Hall and André 2001).

Methodological inconsistencies hinder comparison. Reported weathering rates draw on different protocols for normalizing by surface area, distance from equilibrium, and residence time are not standardized, which reduces interpretability across studies (Weber and Martinez 2017). Hydrologic measurements, including discharge, residence time distributions, and routing to streams, are frequently incomplete despite their central role in translating state changes into fluxes. Expanded use of sensor networks, tracer tests, and remote sensing would improve transport characterization, while coordinated designs can align geochemical sampling with hydrologic state variables (Strefler et al. 2018; Zhang et al. 2022; Calabrese et al. 2022; Baek et al. 2023). Baseline surveys of soils and geology are also uneven, which complicates detection of treatment effects where background inputs are substantial.

7.3 ERW modeling gaps

The data limitations above propagate directly into modeling uncertainty. Several limitations recur across modeling tiers. Hydrologic and biogeochemical feedbacks are often simplified, even though moisture regimes, ion exchange, and biological exudates can alter reaction environments through time and across landscapes (Calabrese et al. 2022). Secondary precipitation and surface passivation are commonly omitted or represented implicitly, which risks overestimating long-term dissolution (Dorn et al. 2013). A persistent gap exists in the coupling between terrestrial models and explicit river-network transport, leaving the fraction of alkalinity that ultimately reaches marine sinks uncertain (Maher and Chamberlain 2014; Sulman et al. 2022). As Zhang et al. (2025) show, neglecting CO_2 degassing in turbulent river systems can significantly inflate projections of oceanic storage, yet few models currently account for this 'leakage' term. Additionally, the reliance on BET (gas adsorption) surface areas in current models can drastically overestimate reaction rates relative to actual geometric or field-effective surface areas, resulting in optimistic projections that fail to materialize in the field (Power et al. 2025).

Land-use dynamics present another gap. Changes in crop type, tillage, irrigation, and harvest practices modify mixing depths, dissolution environments, and logistics, while broader land-cover shifts alter hydrologic connectivity and residence times (Beerling et al. 2025a). Supply chains and logistics are typically treated outside process models, yet source–sink distances, transport modes, and handling constraints influence both emissions and cost trajectories. Recent lifecycle analyses emphasize that comminution and transportation represent the primary environmental hotspots, with trade-offs differing significantly between mined igneous rock and industrial waste streams (Izar-Tenorio et al. 2024). Furthermore, high-resolution spatial modeling has begun to map these logistical constraints against agronomic viability, filtering out theoretical deployment sites that lack proximity to rail networks or possess unsuitable soil buffering capacities (Beerling et al. 2025b). Integrated frameworks that exchange information between process modules and logistics components would yield more realistic scenarios.

Linkages to empirical data can be tightened. Common benchmark datasets from laboratory and field trials would support systematic validation and intercomparison. New initiatives, such as the ERW Field Data Partnership (https://cascadeclimate.org/blog/erw-field-grants) and updated crediting methodologies (https://puro.earth/our-blog/353-puro-earth-launches-enhanced-rock-weathering-methodology-update), are establishing the standardized protocols needed to ground-truth models. Hybrid strategies that employ machine-learning emulators to represent expensive process modules offer a practical route to ensemble projections and uncertainty propagation, provided that emulators are trained within the envelopes defined by experiments (Derry et al. 2025). A modular, multi-scale, data-informed architecture would allow fine-scale kinetics to inform larger-scale predictions while maintaining transparency about uncertainties (Clarkson et al. 2024).

7.4 Environmental safety and socio-economic integration

Beyond technical uncertainties, ERW deployment also faces unresolved environmental safety and socio-economic questions. Long-term environmental effects remain incompletely quantified. Soil chemistry, surface and groundwater quality, and broader ecosystem dynamics can respond in site-specific ways to ERW, and the influence of feedstock variability on these endpoints is not yet well characterized (Edwards et al. 2017; de Oliveira Garcia et al. 2020; Lewis et al. 2021; Eufrasio et al. 2022). There is a specific knowledge gap regarding the 'tipping points' where synergistic benefits turn into antagonistic impacts. For instance, while trace metals like Ni and Cr are inherent to ultramafic rocks, the long-term accumulation rates that trigger toxicity for soil engineers (e.g., earthworms) remain poorly constrained across different soil types (Levy et al. 2024). Defining these safety thresholds is as critical as defining capture potential. Socio-economic implications for small-scale and subsistence farmers are also underexplored. Costs of access, labour demands, and infrastructure gaps may limit adoption, and regional differences in market institutions and resource availability suggest that one policy design will not fit all contexts (Beerling et al. 2020; Val Martin et al. 2023). Community perspectives and cultural factors likely affect acceptance and stewardship but are rarely integrated into study designs.

7.5 Policy bottlenecks and implementation barriers

Lack of clear regulatory frameworks creates uncertainty for investors and implementers, while limited incentives reduce early adoption (Beerling et al. 2020). Public awareness and acceptance vary across settings, which affects siting and permitting (Pidgeon and Spence 2017; Spence et al. 2021). Infrastructure for processing, transport, and application is uneven, and large programs would require coordinated build-out across sectors that do not typically interact (Moosdorf et al. 2014; Tan et al. 2022). Land tenure and land-use priorities can further constrain access, particularly where policies are restrictive or contested (Smith et al. 2019). Equity and environmental safety considerations are important for legitimacy and durability yet are not consistently embedded in policy design.

7.6 Measurement, reporting, and verification for ERW

Despite growing interest in ERW as a scalable CDR method, significant knowledge gaps persist in MRV, undermining accurate quantification and public trust (Clarkson et al. 2024; Beerling et al. 2025a). One major gap involves methodological uncertainties in proxy-based approaches, such as using rare earth element (REE) concentration changes to estimate CDR rates, which may overestimate removal due to unaccounted soil heterogeneity and non-uniform dissolution; critiques highlight that ordinary least squares regression often fails to capture these variabilities, leading to inflated or unreliable results (Kantola et al. 2023; Wolf et al. 2024). Similarly, discrepancies between solid-phase (e.g., soil carbonates), aqueous-phase (e.g., bicarbonate in leachate), and gas-phase MRV methods remain unresolved, with solid methods potentially overlooking transient losses and aqueous methods underestimating due to incomplete sampling (Clarkson et al. 2024). Rock-specific factors, including composition, dissolution sequences, carbonate impurities in feedstocks, and interference from strong acids (e.g., from fertilizers), further complicate verification, as they can decouple cation release from actual CO_2 consumption without standardized corrections (Power et al. 2025). Downstream dynamics introduce additional gaps: dynamic river network models reveal spatial heterogeneity in carbon leakage, where bicarbonate export varies by watershed hydrology, yet integrated basin-scale MRV frameworks are lacking (Zhang et al. 2025). Time lags from cation sorption in soil columns and secondary mineral precipitation also delay verifiable bicarbonate export, creating mismatches between field weathering and long-term sequestration (Beerling et al. 2025a). Bridging these gaps demands interdisciplinary protocols, multi-method integration, and field-scale trials to enhance MRV robustness. For deployment, robustness also requires cost-aware sampling designs that specify which measurements are continuous, which are event-based, and which are reserved for periodic laboratory calibration or third-party verification. Proxies such as pH or electrical conductivity can reduce sampling burden when used to guide

sampling windows and detect anomalies, but they should not replace alkalinity fluxes, cation budgets, and uncertainty treatment in creditable CDR accounting (Stubbs et al. 2022; Clarkson et al. 2024).

8 Recommendations for enhancing the deployment, safety, equity, and scalability of ERW

To improve the deployment and efficacy of ERW as an environmental mitigation strategy, strategic interventions are required. These recommendations focus on addressing monitoring technologies, research frameworks, modeling approaches, environmental safety, policy support, and cross-sector collaboration while ensuring safety, equity, and scalability:

Transition from Proxies to Flux-Based, Cross-scale Monitoring: Correct accounting of ERW CDR in the field requires using pH as one diagnostic variable within flux-based MRV, rather than treating it as direct evidence of removal. Future protocols should prioritize the deployment of sensors capable of quantifying alkalinity fluxes and cation discharge in real-time (Stubbs et al. 2022), and the integration of these sensors with other ground and remote sensors to track hydrological, geological, and ecological changes at various scales (Calabrese et al. 2022). Improved global mapping of mineral resources (Maus et al. 2020) and soil properties (Chen et al. 2022) are essential for identifying safe and effective ERW sites, ensuring that potential environmental risks are minimized. This data will also support the equitable distribution of ERW benefits, as areas with varying geological contexts can be better understood and managed for long-term sustainability.

Establish Large-Scale, Adaptive Research Frameworks: Laboratory columns and short-term plot trials cannot resolve the long-term and large-scale potentials and risks of ERW. Research frameworks should enable long-term data collection under diverse environments, and combine geospatial analysis with comprehensive soil, hydrological, and geological surveys to identify key regions with scalable ERW potential (Beerling et al. 2020). Research in croplands should focus on agronomic integration and food safety (e.g., trace-metal risks), while pilots in natural ecosystems (e.g., forests) must prioritize the monitoring of deep-soil inorganic carbon storage and biodiversity conservation, acknowledging their fundamentally different implementation pathways. The introduction of pilot projects in diverse ecological and agricultural settings will provide critical, context-specific data on ERW's impacts, ensuring safe practices and equitable access to its benefits (Levy et al. 2024). Additionally, focusing on small-scale and subsistence farming communities will help ensure that ERW strategies are not only scalable but also accessible and fair (Haque et al. 2021).

Integrate Holistic and Inclusive Modeling Approaches: Useful ERW models need to link mineral dissolution kinetics to hydrology and soil carbon cycling, which requires representing reactive transport with effective surface area evolutions and microbial-plant-soil feedbacks within a single framework. Integrated assessment efforts should use realistic field CDR rates generated by the ERW models, while incorporating land-use changes (Zeng et al. 2022), logistical challenges (Renforth 2012; Tan et al. 2022), and socio-economic factors (Strefler et al. 2018) to ensure that ERW practices are scalable and equitable (Tan et al. 2022).

Assess and Mitigate Environmental Risks: Any large-scale deployment of crushed mafic or ultramafic rock into the soil system must come after site-specific evaluation of trace-metal and mineralogical risk profiles that vary with the rock type and the geosocial and ecological contexts (Lawford-Smith and Currie 2017; Vandeginste et al. 2024). Localized, robust risk mitigation strategies should be developed to proactively address any negative consequences on soil health, food safety, water quality, and ecosystems. Ensuring that these strategies are equitable and do not disproportionately impact vulnerable regions or communities is critical for the responsible and sustainable scaling of ERW (Oppon et al. 2024a).

Implement Comprehensive Policy and Regulatory Support: Governments and international organizations should develop clear regulatory frameworks and provide financial incentives to support safe and equitable ERW deployment (Webb 2020). This includes implementing MRV systems that ensure ERW projects adhere to safety standards and contribute effectively to carbon sequestration goals (Smith et al. 2020). Policymaking should prioritize public awareness, infrastructure development, and international collaboration to foster ERW's global scalability (Spence et al. 2021). Special attention should

be given to the inclusion of marginalized and under-resourced communities in these frameworks, ensuring equitable access to the benefits of ERW.

Strengthen Academia-Industry Collaboration: Academic institutions and industry should partner to accelerate innovation and ensure that research aligns with the practical needs of ERW deployment, particularly regarding safety and scalability (Power et al. 2023). Collaborative platforms should facilitate the exchange of data, insights, and best practices, reducing the time lag between academic discoveries and field implementation (Spence et al. 2021). Securing policy support and financial incentives for research and pilot projects, alongside addressing logistical challenges, will be key to scaling ERW while ensuring that the benefits are distributed equitably and safely across different sectors and regions (Cox et al. 2022).

9 Concluding remarks

ERW occupies a promising, yet contingent, position within the global carbon removal portfolio. Its capacity to pair alkalinity generation with tangible co-benefits for terrestrial ecosystems creates opportunities in settings where specific soils and hydrology support the export of alkalinity to durable sinks. However, realizing this potential requires a paradigm shift from theoretical potential to verifiable performance. The mineralogical and ecohydrological heterogeneity and modeling uncertainties highlighted in this review demonstrate that fast kinetic constants observed in laboratory settings do not guarantee effective carbon removal in the field. A practical path forward requires linking site selection to rigorous measurements of alkalinity and cation fluxes, together with soil and vegetation characteristics, while incorporating careful risk assessments, particularly for trace metal accumulation. It is essential to develop data-informed models that capture field weathering complexities and account for downstream riverine and ocean losses. Lifecycle effectiveness, environmental safety, and distributional considerations will ultimately shape public acceptance and policy support, particularly where feedstock chemistry, dust, and land use interact with local priorities. When managed under these conditions, ERW can contribute substantially alongside other approaches to carbon removal and soil restoration. Ultimately, ERW's impact on environmental mitigation will not be defined by the vastness of available rocks, but by our ability to safely deploy these minerals within planetary boundaries and credibly measure the net carbon sequestered at geological timescales.

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Data availability The authors declare that no new data were used for created in producing this manuscript. Figures 1 and 4 were created using Adobe Photoshop and Adobe Illustrator. Figure 2 was created using PGF (a graphing package in latex). Dataset used to create Fig. 2 can be directly derived from Table 3 of this paper, and Bandstra and Brantley (2008). Figures 3 was created using Canva and is based on data from Table S1.

Declarations

Conflict of interest The authors declare no conflicts of interest relevant to this study.

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